# PSYC 4 Midterm 2 Notes（Ch 6 to 10）

**作答規則（模擬考二的陷阱）**
1. 先讀最後一句問句，遮住選項，心裡說答案。
2. 圈出題目說 normal、intact、still、can、full strength、recognizes 的地方，先劃掉相關選項。
3. 看到 NOT、EXCEPT、goes beyond，題號旁寫大 **F**：要找錯的那一句。
4. **選項裡有下面的成對概念，就算確定也標？，翻這頁核對方向。**
5. 每個選項整句讀完，找到一個錯字才劃掉；半對半錯＝錯。
6. 受器的第一步是機械性拉開（mechanical），選項寫 voltage first 就劃掉。
7. **第一個答案留著，找到字證明它錯才改。**

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| **Rods 桿細胞**<br>周邊、暗光敏感、看不到顏色、多個接一個雙極細胞（模糊） | **Cones 錐細胞**<br>中央小窩、要亮光、看顏色、一對一（清楚） |
| **Photoreceptors in the DARK 黑暗**<br>去極化、cGMP 高、鈉通道開、一直放 glutamate、壓住雙極細胞 | **Photoreceptors in LIGHT 有光**<br>cGMP 被分解、鈉通道關、過極化、glutamate 變少、雙極細胞被放開 |
| **Wavelength 波長**<br>顏色 hue（光）／音高 pitch（聲音＝frequency） | **Amplitude 振幅**<br>亮度 brightness（光）／響度 loudness（聲音） |
| **Iris 虹膜**<br>肌肉，窗簾，控制瞳孔大小 | **Pupil 瞳孔**<br>只是一個洞，什麼都不控制 |
| **Fovea 中央小窩**<br>錐細胞最多，看得最清楚 | **Optic disc 視神經盤**<br>軸突離開的後門，沒有感光細胞＝盲點 |
| **Trichromatic 三色說**<br>錐細胞那一層（三種錐細胞混色） | **Opponent-process 拮抗說**<br>神經節細胞那一層（紅綠、藍黃、黑白；後像） |
| **Ventral stream 腹側**<br>顳葉，WHAT：認東西、認臉（壞了＝臉盲 prosopagnosia） | **Dorsal stream 背側**<br>頂葉，WHERE／動態：MT 壞了＝看不到動 akinetopsia |
| **Simple cell**<br>固定角度＋固定位置 | **Complex cell**<br>固定角度、位置不限，喜歡會動的 |
| **Base 基底膜底部**<br>窄、硬、靠近 oval window：高音；噪音先打壞這裡 | **Apex 頂端**<br>寬、軟：低音 |
| **Touch 觸覺路線**<br>同側往上，在延腦 medulla 交叉 | **Pain 痛覺路線**<br>一進脊髓馬上交叉 |
| **脊髓單側受傷**<br>傷哪邊，哪邊沒觸覺 | <br>痛覺沒的是另一邊 |
| **Mechanoreceptors**<br>壓力、震動、拉長（Pacinian：壓洋蔥、鈉進去） | **Nociceptors**<br>痛：傷害、極端溫度、化學物質 |
| **Gate：短暫緊急**<br>把痛的門關上（受傷沒感覺） | **Gate：長期壓力、缺睡、餓**<br>門打開（更痛） |
| **Smell 嗅覺**<br>跳過視丘，直通杏仁核、海馬迴 | **Taste 味覺**<br>延腦 → 視丘 → 味覺皮質 |

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| **Muscle spindle 肌梭**<br>偵測拉長 length；膝跳反射，單突觸 | **Golgi tendon organ**<br>偵測張力 tension；抑制性中間神經元，讓肌肉放鬆 |
| **Monosynaptic 單突觸**<br>膝跳反射，最快 | **Polysynaptic 多突觸**<br>縮手反射（痛覺受器＋中間神經元） |
| **Dorsal 背側**<br>感覺進來 IN | **Ventral 腹側**<br>運動出去 OUT（alpha 運動神經元在腹角） |
| **Corticospinal 交叉點**<br>延腦和脊髓交界處 | **終點**<br>脊髓腹角（接 alpha 運動神經元） |
| **Acetylcholine**<br>神經肌肉接合處 → 鈉進肌肉 → 收縮 | **Dopamine**<br>基底核油門（黑質），也管快樂 |
| **Gas 油門＝substantia nigra**<br>多巴胺不夠＝帕金森：靜止會抖、很難開始動、先憂鬱、先單側 | **Brakes 煞車＝striatum**<br>壞了＝杭丁頓：停不下來的抽動、顯性基因 |
| **FES**<br>讀出訊號 OUT，繞過脊髓，刺激肌肉 | **DBS**<br>送電刺激 IN 到大腦（帕金森） |
| **Drive reduction 驅力降低**<br>推 push：補足缺乏（餓了吃餅乾） | **Reward 獎賞**<br>拉 pull：誘因（飽了還吃蛋糕） |
| **Homeostasis**<br>概念、過程 | **Set point**<br>具體的設定值（例如 37 度） |
| **Endotherm 內溫**<br>自己產熱（松鼠、人） | **Ectotherm 外溫**<br>靠環境（蜥蜴曬太陽） |
| **Hyperthermia**<br>體溫太高（hyper 太多） | **Hypothermia**<br>體溫太低（hypo 太少） |
| **Hypotonic 低張**<br>外面稀 → 水跑進細胞 → 脹大 | **Hypertonic 高張**<br>外面濃（吃鹹、3% 鹽水）→ 水跑出來 → 縮水 |
| **Osmotic thirst 滲透壓口渴**<br>吃鹹，細胞縮水，OVLT 偵測；補水就好 | **Hypovolemic thirst 低血容量口渴**<br>失血、拉肚子，baroreceptors、腎臟、angiotensin II → subfornical organ；要補水＋鹽 |
| **Insulin 胰島素**<br>血糖高時：葡萄糖進細胞、存成肝醣 | **Glucagon 升糖素**<br>血糖低時：肝醣變回葡萄糖 |
| **Ghrelin**<br>空腹的胃 → 弓狀核 → orexin → 想吃 | **Leptin**<br>脂肪說「存夠了」；胖的人常有阻抗 |
| **Agonist 促效劑**<br>讓神經傳導物質作用變強（古柯鹼擋回收） | **Antagonist 拮抗劑**<br>擋住受體，作用變弱 |
| **MIH**<br>殺掉 Müllerian（女性管道） | **Testosterone**<br>養 Wolffian（男性管道） |
| **DHT（要 5-alpha reductase）**<br>外生殖器：陰莖 | **Estradiol（要 aromatase）**<br>大腦男性化 |
| **Organizational 塑造**<br>出生前荷爾蒙，永久改變大腦（CAH） | **Activational 啟動**<br>青春期以後荷爾蒙，啟動行為 |
| **外來睪固酮**<br>負回饋：壓低 GnRH、LH、FSH → 睪丸萎縮 | <br>不是調高 |

# Study Guide 單字表：情境裡看到這些，答案就是它

| 第 6 章 | | |
|---|---|---|
| **Sensation**<br>sen-SAY-shun | 感覺：收訊號<br>receiving signals and turning them into neural signals | **eyes or ears test normal; signals are received**<br>眼睛、耳朵正常、收得到 |
| **Perception**<br>per-SEP-shun | 知覺：大腦詮釋<br>the brain organizing and interpreting signals | **receives signals but cannot recognize; two people see the same image differently**<br>收得到但認不出、兩個人看同一張圖看法不同 |
| **Bottom-up processing**<br>BOT-um UP | 由下而上<br>building meaning from small details | **first time seeing it; no expectations; reading letter by letter**<br>第一次看到、沒有預期、一個字母一個字母讀 |
| **Top-down processing**<br>TOP DOWN | 由上而下<br>using knowledge and expectations | **expectations, familiarity, missing a typo, stereotypes**<br>期待、熟悉、沒看到錯字、刻板印象 |
| **Photon**<br>FOH-ton | 光子<br>a particle of light | **more photons = brighter**<br>光子變多＝更亮 |
| **Wavelength**<br>WAVE-length | 波長<br>distance between waves; codes color | **color; red vs green light; 530 nm**<br>顏色、紅光綠光、530 nm |
| **Amplitude**<br>AM-plih-tood | 振幅<br>height of the wave; codes brightness or loudness | **brighter, dimmer, louder, taller wave**<br>更亮、更暗、更大聲、波變高 |
| **Refraction**<br>rih-FRAK-shun | 折射<br>bending of light | **blurry underwater; goggles; focusing at the cornea**<br>水裡看不清、蛙鏡、角膜、對焦 |
| **Cornea**<br>KOR-nee-uh | 角膜<br>clear front surface; does most bending | **where light enters from the air**<br>光從空氣進眼睛那一層 |
| **Lens**<br>LENZ | 水晶體<br>adjusts shape to focus near and far | **thickens to see near objects; fails with age**<br>近看要變厚、老花 |
| **Iris**<br>EYE-ris | 虹膜<br>colored muscle that sets pupil size | **pupil gets bigger or smaller; eye drops**<br>瞳孔變大變小、眼藥水 |
| **Pupil**<br>PYOO-pil | 瞳孔<br>the opening; controls nothing | **just an opening**<br>只是洞 |
| **Fovea**<br>FOH-vee-uh | 中央小窩<br>center of retina, packed with cones | **looking straight at it; sharpest; color**<br>正對著看、最清楚、顏色 |

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| **Optic disc**<br>OP-tik DISK | 視神經盤<br>where axons leave; no receptors | **blind spot; a dot disappears**<br>盲點、點消失 |
| **Rods**<br>RODZ | 桿細胞<br>sensitive, dim light, no color, periphery | **dim light; corner of the eye; look to the side at a faint star**<br>暗處、眼角、星星往旁邊看 |
| **Cones**<br>KOHNZ | 錐細胞<br>bright light, color, detail, fovea | **color, detail, needs bright light**<br>顏色、細節、要亮 |
| **Bipolar cells**<br>by-POH-ler | 雙極細胞<br>middle layer; released when light arrives | **held down in the dark; fire when light arrives**<br>黑暗中被壓住、光來才發 |
| **Ganglion cells**<br>GANG-glee-on | 神經節細胞<br>last layer; first to fire action potentials; axons form optic nerve | **last layer; opponent pairs; leave through the optic disc**<br>最後一層、拮抗配對、走出視神經盤 |
| **Horizontal / amacrine cells**<br>hor-ih-ZON-tul / AM-uh-krin | 水平／無軸突細胞<br>side connections; lateral inhibition | **side connections; sharper edges**<br>側向連接、邊緣變清楚 |
| **Glutamate (in the dark)**<br>GLOO-tuh-mate | 麩胺酸<br>released by photoreceptors in the dark to inhibit bipolar cells | **released all the time in the dark; less in light**<br>黑暗中一直放、光來變少 |
| **cGMP**<br>see-jee-em-PEE | cGMP<br>keeps sodium channels open in the dark; light breaks it down | **not broken down = acts like darkness**<br>不被分解＝一直像黑暗 |
| **Receptive field (center-surround)**<br>ree-SEP-tiv FEELD | 感受野<br>area a cell responds to; center excites, surround inhibits | **center light strong; surround light weak; whole field cancels**<br>照中心強、照外圈弱、整片抵消 |
| **Lateral inhibition**<br>LAT-er-al in-hih-BISH-un | 側向抑制<br>neighbors inhibit each other to sharpen edges | **edges, contrast, Hermann grid**<br>邊緣、對比、Hermann grid |
| **Trichromatic theory**<br>try-kroh-MAT-ik | 三色說<br>three cone types mix colors | **cones; color mixing; red + green = yellow**<br>錐細胞、混色、紅＋綠＝黃 |
| **Opponent-process theory**<br>oh-POH-nent | 拮抗說<br>red-green, blue-yellow pairs in ganglion cells | **afterimage; stare, then see the opposite color**<br>後像、盯久了看到相反色 |
| **Simple / complex cells**<br>SIM-pul / KOM-plex | 簡單／複雜細胞<br>V1 cells: fixed spot vs anywhere, likes motion | **fixed spot vs anywhere and moving**<br>固定位置 vs 哪裡都行會動 |
| **Ventral stream**<br>VEN-trul | 腹側路線<br>to temporal lobe: WHAT (objects, faces); damage = prosopagnosia (PROS-oh-pag-NOH-zhuh), face blindness | **recognize objects and faces; face blindness**<br>認東西、認臉、臉盲 |

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| **Dorsal stream**<br>DOR-sul | 背側路線<br>to parietal lobe: WHERE and motion (area MT); damage = akinetopsia (ay-kin-eh-TOP-see-uh), motion blindness | **cannot see motion; frozen snapshots**<br>看不到動、像照片一格格 |
| **第 7 章** | | |
| **Frequency**<br>FREE-kwen-see | 頻率<br>cycles per second; codes pitch | **pitch; cycles per second**<br>音高、每秒幾次 |
| **Basilar membrane**<br>BAZ-ih-ler | 基底膜<br>in cochlea; base = high pitch, apex = low pitch | **cannot hear high pitch = base damaged**<br>高音聽不到＝底部壞 |
| **Hair cells / stereocilia**<br>steer-ee-oh-SIL-ee-uh | 毛細胞／靜纖毛<br>mechanoreceptors; bending pulls channels open | **bending pulls channels open mechanically, not voltage**<br>毛彎曲、機械性拉開、不是電壓 |
| **Medial geniculate nucleus (MGN)**<br>jeh-NIK-yoo-lit | 內側膝狀核<br>thalamus relay for hearing | **hearing relay in the thalamus**<br>聽覺經視丘 |
| **Auditory cortex**<br>AW-dih-tor-ee | 聽覺皮質<br>temporal lobe | **temporal lobe**<br>顳葉 |
| **McGurk effect**<br>mik-GURK | McGurk 效應<br>seeing lips changes what you hear | **sound ba + lips ga = hear da; eyes closed restores ba**<br>聲音 ba 嘴型 ga 聽到 da、閉眼恢復 |
| **Association cortex**<br>uh-soh-see-AY-shun | 聯合皮質<br>areas that combine senses | **combines vision with hearing or touch**<br>整合視覺和聽覺／觸覺 |
| **Somatosensory cortex**<br>soh-mat-oh-SEN-sor-ee | 體感皮質<br>postcentral gyrus; receives touch | **tingling; feeling; behind the central sulcus**<br>刺刺的、感覺、中央溝後面 |
| **Motor cortex**<br>MOH-ter | 運動皮質<br>precentral gyrus; sends movement commands | **twitching; sending commands; in front of the central sulcus**<br>抽動、下命令、中央溝前面 |
| **Mechanoreceptors**<br>meh-KAN-oh-ree-sep-terz | 機械受器<br>respond to pressure, vibration, stretch | **pressure, vibration, Pacinian, hair cells**<br>壓、震動、Pacinian、毛細胞 |
| **Pacinian corpuscle**<br>pah-SIN-ee-un KOR-pus-ul | Pacinian 小體<br>onion-shaped; sudden pressure and vibration | **responds to sudden pressure, ignores steady pressure**<br>突然壓才有反應、持續壓不理 |
| **Thermoreceptors**<br>THER-moh-ree-sep-terz | 溫度受器<br>respond to warm or cold | **warm or cold (not pain)**<br>冷熱（不是痛） |
| **Nociceptors**<br>NOH-sih-sep-terz | 痛覺受器<br>respond to damage, extreme heat, chemicals | **injuries not felt; normal axons and myelin but no pain**<br>受傷沒感覺、軸突髓鞘正常卻不痛 |

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| **Proprioceptors**<br>proh-pree-oh-SEP-terz | 本體感受器<br>in muscles; length and tension | **knows hand position with eyes closed; cannot tell where feet are without looking**<br>閉眼知道手在哪、不看不知道腳在哪 |
| **Vestibular sense**<br>ves-TIB-yoo-ler | 前庭覺<br>balance; fluid in semicircular canals | **dizzy after spinning; balance**<br>轉圈頭暈、平衡 |
| **Two-point discrimination**<br>dis-krim-ih-NAY-shun | 兩點辨識<br>telling two touches apart | **two points on fingertip, one on upper arm**<br>指尖兩點、上臂一點 |
| **Dermatome**<br>DER-muh-tome | 皮節<br>skin area served by one spinal level | **no feeling below a certain level**<br>某個高度以下沒感覺 |
| **Dorsal column pathway**<br>DOR-sul KOL-um | 觸覺路線<br>touch; same side, crosses in medulla | **touch lost on the same side**<br>同側失去觸覺 |
| **Spinothalamic pathway**<br>spy-noh-thuh-LAM-ik | 痛覺路線<br>pain; crosses right away in cord | **pain lost on the opposite side**<br>對側失去痛覺 |
| **Homunculus**<br>hoh-MUNK-yoo-lus | 小人圖<br>body map; hands, lips large; face next to hand | **area depends on fine control; touching the face feels like the missing hand**<br>面積看精細度、摸臉覺得手被碰 |
| **Plasticity**<br>plas-TIS-ih-tee | 可塑性<br>map changes with use; slow, not instant | **violinist; weeks after amputation (not seconds)**<br>小提琴家、截肢後幾週（不是幾秒） |
| **Phantom limb pain**<br>FAN-tum LIM | 幻肢痛<br>pain in a missing limb | **amputee; still clenching**<br>截肢、還在握拳 |
| **Mirror box**<br>MIR-er BOX | 鏡箱<br>vision supplies the missing feedback | **seeing the hand open releases the cramp**<br>看到手張開就鬆開 |
| **Gate theory**<br>GATE | 閘門理論<br>touch input closes a spinal gate on pain | **rubbing helps; stress and sleep loss make it worse**<br>揉一揉、壓力缺睡會更痛 |
| **Olfactory bulb**<br>ol-FAK-tor-ee | 嗅球<br>smell; skips thalamus, to amygdala and hippocampus | **a smell brings back old memories**<br>聞到就想起往事 |
| **Gustatory pathway**<br>GUS-tuh-tor-ee | 味覺路線<br>taste: medulla, thalamus, gustatory cortex | **taste goes through the thalamus**<br>味覺經視丘 |
| **Insula**<br>IN-suh-luh | 腦島<br>where taste, smell, touch combine into flavor | **pinched nose leaves only sweet**<br>捏鼻子只剩甜 |
| **第 8 章** | | |
| **Muscle spindle**<br>SPIN-dul | 肌梭<br>detects stretch (length) | **knee jerk; muscle stretched**<br>膝跳、被拉長 |

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| **Golgi tendon organ**<br>GOAL-jee TEN-dun | 高爾基腱器<br>detects tension (force); relaxes muscle | **drops a heavy object; too much force makes the muscle relax**<br>太重放手、用力太大鬆開 |
| **Alpha motor neuron**<br>AL-fuh | alpha 運動神經元<br>in ventral horn; makes muscle contract | **ventral horn; last nerve to the muscle**<br>腹角、最後一條神經 |
| **Acetylcholine (ACh)**<br>uh-see-til-KOH-leen | 乙醯膽鹼<br>neuromuscular junction transmitter | **muscles do not contract although the nerves are normal**<br>肌肉不收縮但神經正常 |
| **Decussation**<br>dek-uh-SAY-shun | 交叉<br>crossing; corticospinal at medulla-cord junction | **asks where the axons cross**<br>問在哪裡 cross |
| **Monosynaptic reflex**<br>mon-oh-sih-NAP-tik | 單突觸反射<br>one synapse; knee jerk | **tapping the knee**<br>敲膝蓋 |
| **Polysynaptic reflex**<br>pol-ee-sih-NAP-tik | 多突觸反射<br>with interneurons; withdrawal | **touching something hot and pulling away**<br>碰燙縮手 |
| **Reciprocal inhibition**<br>rih-SIP-roh-kul | 相互抑制<br>opposite muscle relaxes | **biceps contracts, triceps relaxes**<br>二頭肌收、三頭肌鬆 |
| **Somatotopic map**<br>soh-mat-oh-TOP-ik | 體部位地圖<br>body map in motor cortex | **fingers and lips take a lot of space**<br>手指嘴唇佔很多 |
| **FES**<br>EF-EE-ES | 功能性電刺激<br>reads cortex signals out, stimulates muscles, bypasses cord | **paralyzed patient uses an implant to grab a cup**<br>四肢癱瘓靠晶片拿杯子 |
| **Cerebellum**<br>sair-uh-BEL-um | 小腦<br>compares intended vs actual; corrects errors | **overshoots, oscillates, cannot adapt to prisms, sobriety test**<br>伸過頭、來回晃、稜鏡適應不了、酒測 |
| **Substantia nigra**<br>sub-STAN-shuh NY-gruh | 黑質<br>dopamine; the GAS | **Parkinson; depression comes first**<br>帕金森、先憂鬱 |
| **Striatum (caudate, putamen)**<br>stry-AY-tum | 紋狀體<br>the BRAKES | **Huntington; caudate and putamen shrink**<br>杭丁頓、尾核殼核萎縮 |
| **Parkinson disease**<br>PAR-kin-sun | 帕金森<br>not enough gas: rest tremor, hard to start moving | **rest tremor; freezes in doorways**<br>靜止會抖、門口卡住 |
| **Huntington disease**<br>HUN-ting-tun | 杭丁頓<br>broken brakes: jerky movements, dominant gene | **jerky movements that will not stop; a parent had it**<br>停不下來的抽動、父母也有 |
| **Demyelination**<br>dee-my-uh-lih-NAY-shun | 髓鞘流失<br>myelin lost; slow, scattered signals | **cortex and muscle normal; signals slow and out of step**<br>皮質肌肉正常、訊號慢又散 |

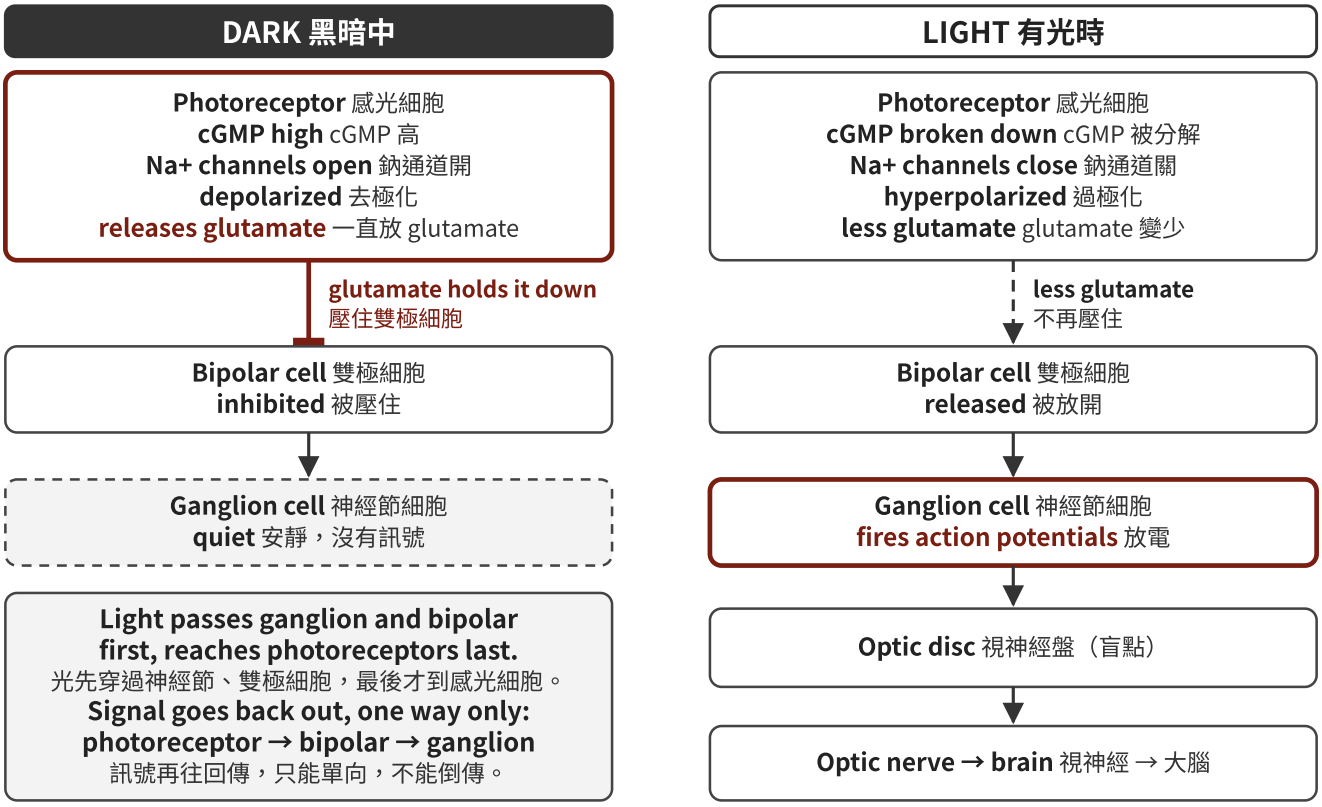
| | | |
|---|---|---|
| **Mirror neurons**<br>MIR-er NOOR-onz | 鏡像神經元<br>fire when doing and watching | **watching someone else act; imitation**<br>坐著看別人做、模仿 |
| **第 9 章** | | |
| **Drive reduction**<br>DRIVE rih-DUK-shun | 驅力降低<br>push: removing a need | **hungry, eats until not hungry**<br>餓了吃到不餓 |
| **Reward / incentive**<br>rih-WARD / in-SEN-tiv | 獎賞<br>pull: wanting something pleasant | **full but still orders cake**<br>飽了還吃蛋糕 |
| **Homeostasis**<br>hoh-mee-oh-STAY-sis | 恆定<br>the concept of keeping balance | **keeping things stable (the concept)**<br>維持穩定（概念） |
| **Set point**<br>SET POINT | 設定值<br>the specific target value | **37 degrees; raised in fever**<br>37 度、發燒調高 |
| **Endotherm**<br>EN-doh-therm | 內溫動物<br>makes own heat | **squirrel, human**<br>松鼠、人 |
| **Ectotherm**<br>EK-toh-therm | 外溫動物<br>heat from environment | **lizard basking in the sun**<br>蜥蜴曬太陽 |
| **Hyperthermia / hypothermia**<br>hy-per-THER-mee-uh / hy-poh- | 體溫過高／過低<br>too hot / too cold | **hyper = too much, hypo = too little**<br>hyper 多、hypo 少 |
| **Preoptic area (POA)**<br>pree-OP-tik | 視前區<br>hypothalamic warm/cold sensing neurons | **fever, shivering, sweating**<br>發燒、發抖、流汗 |
| **Pyrogens**<br>PY-roh-jenz | 致熱原<br>raise the set point (fever) | **normal temperature but shivering**<br>正常體溫卻發抖 |
| **Isotonic**<br>eye-soh-TON-ik | 等張<br>equal concentration; no net water movement | **no change**<br>不變 |
| **Hypotonic**<br>hy-poh-TON-ik | 低張<br>outside more dilute; water enters; cell swells | **very dilute IV; drinking too much water**<br>很稀的點滴、喝太多水 |
| **Hypertonic**<br>hy-per-TON-ik | 高張<br>outside more concentrated; water leaves; cell shrinks | **salty food; 3% saline**<br>吃鹹、3% 鹽水 |
| **Osmotic thirst**<br>oz-MOT-ik | 滲透壓口渴<br>cells shrink; OVLT osmoreceptors | **salty food; no blood loss**<br>吃鹹、沒失血 |
| **Hypovolemic thirst**<br>hy-poh-voh-LEE-mik | 低血容量口渴<br>blood volume drops; baroreceptors, angiotensin II to subfornical organ | **blood loss, diarrhea; needs salt too**<br>失血、拉肚子；要補鹽 |
| **ADH (vasopressin)**<br>AY-DEE-AYCH | 抗利尿激素<br>kidneys save water | **saves water; less urine**<br>省水、尿變少 |

| | | |
|---|---|---|
| **Glucoreceptors**<br>gloo-koh-ree-SEP-terz | 葡萄糖受器<br>detect blood glucose | **detect blood glucose**<br>偵測血糖 |
| **Insulin**<br>IN-suh-lin | 胰島素<br>high sugar: store it | **right after a meal**<br>剛吃飽 |
| **Glucagon**<br>GLOO-kuh-gon | 升糖素<br>low sugar: release it from glycogen | **long fasting; marathon**<br>餓很久、馬拉松 |
| **Ghrelin**<br>GREL-in | 飢餓素<br>stomach hunger signal | **empty stomach; wants to eat**<br>空腹、想吃 |
| **Leptin**<br>LEP-tin | 瘦素<br>fat says stores are full; resistance in obesity | **obese person with high leptin is still hungry**<br>胖的人 leptin 高還是餓 |
| **Orexins**<br>oh-REK-sinz | 食慾素<br>lateral hypothalamus; promote eating | **start eating**<br>啟動進食 |
| **Arcuate nucleus**<br>AR-kyoo-it | 弓狀核<br>hypothalamus hub for hunger signals | **where ghrelin and leptin act**<br>ghrelin、leptin 作用的地方 |
| **VTA and nucleus accumbens**<br>NOO-klee-us uh-KUM-benz | 獎賞迴路<br>dopamine reward circuit | **pleasure, addiction**<br>快樂、成癮 |
| **Cocaine**<br>koh-KANE | 古柯鹼<br>blocks dopamine reuptake: agonist | **blocks the transporter = stronger effect**<br>擋轉運體＝作用變強 |
| **第 10 章** | | |
| **SRY / TDF**<br>ES-AR-WHY | SRY／睪丸決定因子<br>Y gene that makes testes | **the first domino**<br>第一張骨牌 |
| **Wolffian ducts**<br>WOOL-fee-un | 沃爾夫管<br>male ducts; need testosterone | **grown by testosterone**<br>睪固酮養 |
| **Müllerian ducts**<br>myoo-LAIR-ee-un | 繆勒管<br>female ducts; default | **killed by MIH**<br>MIH 殺掉 |
| **MIH**<br>EM-EYE-AYCH | 繆勒管抑制激素（課本叫 AMH）<br>from testes; kills Müllerian (textbook calls it AMH) | **present whenever there are testes**<br>有睪丸就有 |
| **DHT / 5-alpha reductase**<br>DEE-AYCH-TEE | 雙氫睪固酮<br>makes penis | **missing = no penis, otherwise male**<br>缺了＝沒陰莖、其他男性 |
| **Aromatase**<br>uh-ROH-muh-tase | 芳香酶<br>turns testosterone into estradiol in neurons | **missing = female brain**<br>沒有＝大腦女性 |
| **Alpha-fetoprotein**<br>AL-fuh fee-toh-PROH-teen | 甲型胎兒蛋白<br>blocks mother's estrogen from fetal brain | **why the mother's estrogen has no effect**<br>媽媽雌激素為什麼不影響 |
| **Sexual dimorphism**<br>dy-MOR-fizm | 性別雙型<br>structures differing by sex | **INAH-3, SDN**<br>INAH-3、SDN |

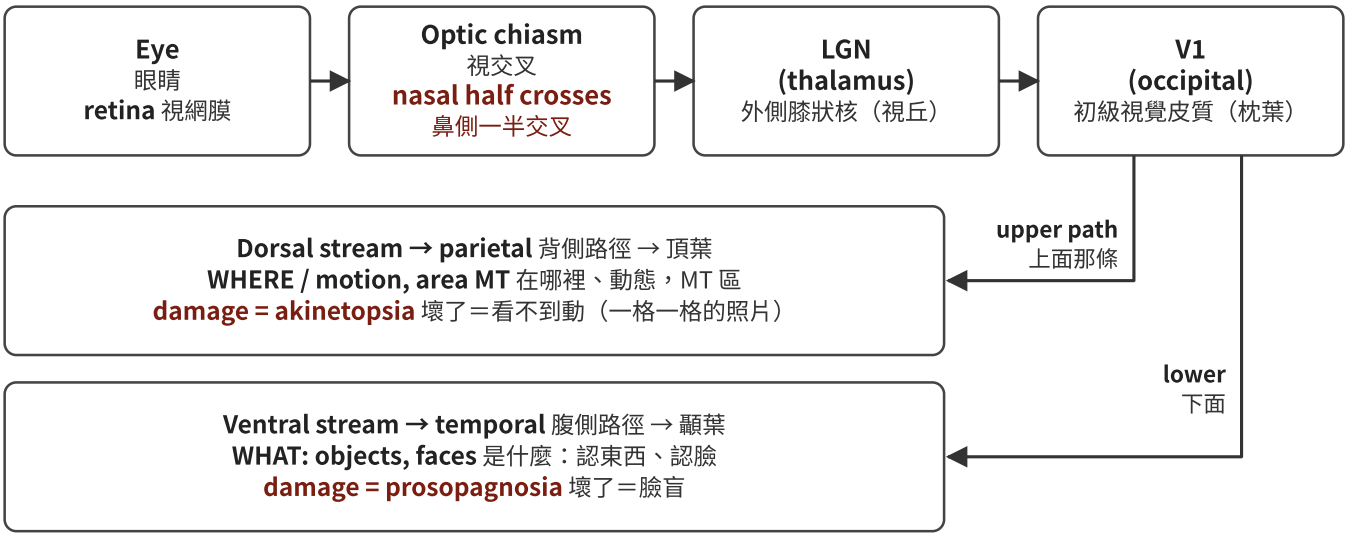
| | | |
|---|---|---|
| **AIS**<br>AY-EYE-ES | 雄性素不敏感<br>receptors don't respond to androgens | **XY with female appearance and no ducts**<br>XY 外表女性、沒管道 |
| **PMDS**<br>PEE-EM-DEE-ES | 持續繆勒管症候群<br>no working MIH | **both duct systems**<br>兩套管道都有 |
| **Klinefelter (XXY)**<br>KLINE-fel-ter | 克氏症<br>male, low testosterone | **tall, small testes**<br>高個子、睪丸小 |
| **Turner (XO)**<br>TUR-ner | 透納氏症<br>female, missing one X | **short, no puberty**<br>矮、沒有青春期 |
| **CAH**<br>SEE-AY-AYCH | 先天性腎上腺增生<br>high prenatal androgens in XX | **girl prefers rough play**<br>女孩喜歡粗野遊戲 |
| **Organizational effects**<br>or-guh-nih-ZAY-shun-ul | 塑造作用<br>prenatal, permanent brain shaping | **before birth; shows up in childhood**<br>出生前、童年就出現 |
| **Activational effects**<br>ak-tih-VAY-shun-ul | 啟動作用<br>later hormones switch behavior on | **after puberty**<br>青春期以後 |
| **GnRH, LH, FSH**<br>jee-EN-AR-AYCH | 性腺軸<br>hypothalamus-pituitary chain to gonads | **outside testosterone = levels drop**<br>外來睪固酮＝壓低 |
| **Oxytocin / vasopressin**<br>ok-see-TOH-sin | 催產素／血管加壓素<br>pair bonding, parenting | **vole pair bonding**<br>田鼠配對 |

# 流程圖：看圖就知道循環

## 1. SG4｜視網膜：黑暗 vs 有光 Retina in DARK vs LIGHT （p.180 to 194, 205 to 206）

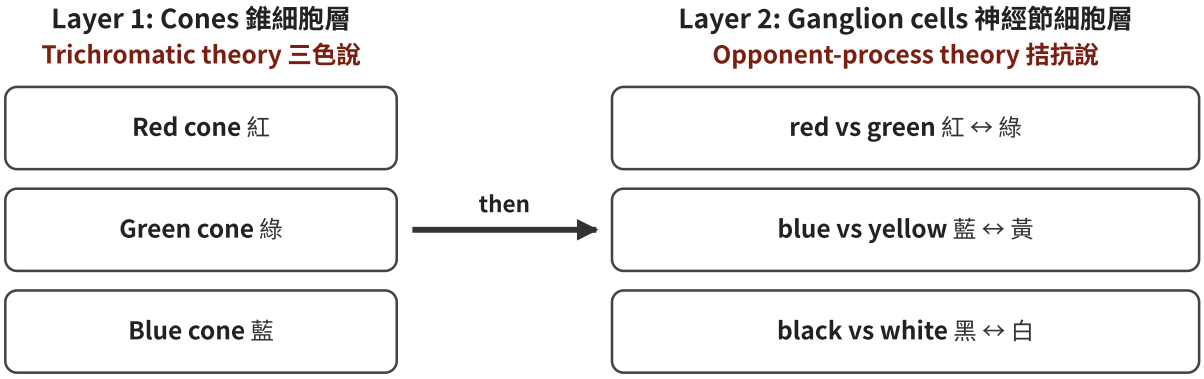



## 2. SG4｜視覺路徑 Visual pathway （p.180 to 194, 205 to 206）



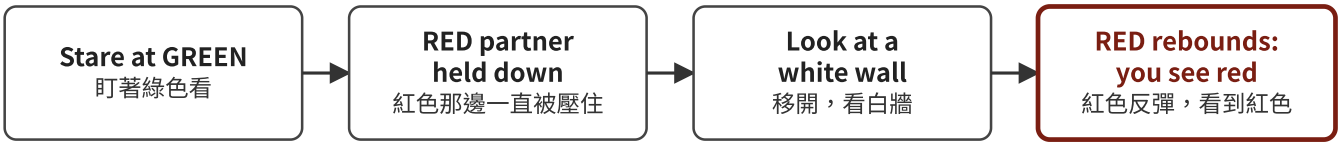

**Still recognizes FACES = ventral is normal: cross out face blindness.**
**Recognizes objects but NOT faces = that IS prosopagnosia.**
認得出「臉」才能劃掉臉盲。認得出東西、卻認不出臉＝正是臉盲。

## 3. SG4｜顏色兩層 Color: trichromatic and opponent-process （p.180 to 194, 205 to 206）



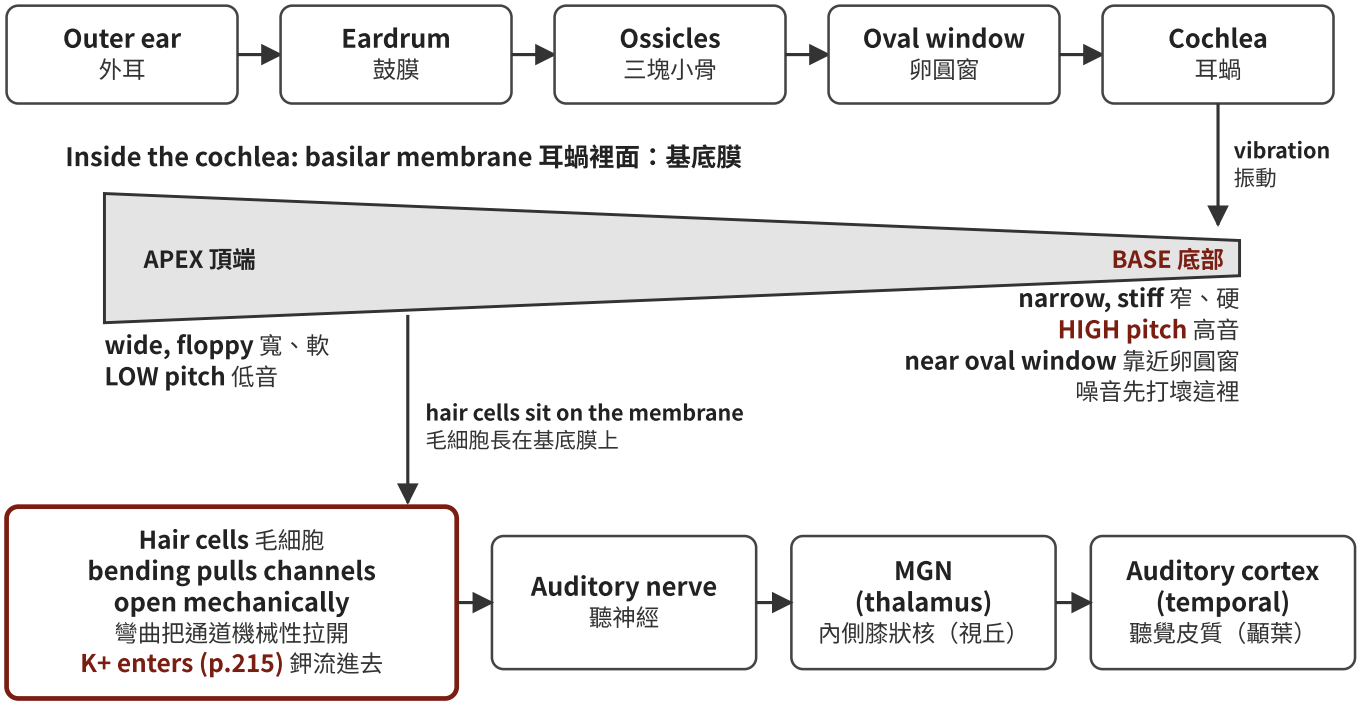

**Both theories are right, at different layers. 兩個都對，只是不同層。**

**Afterimage 後像: the suppressed partner rebounds 一直被壓住的那一邊反彈**



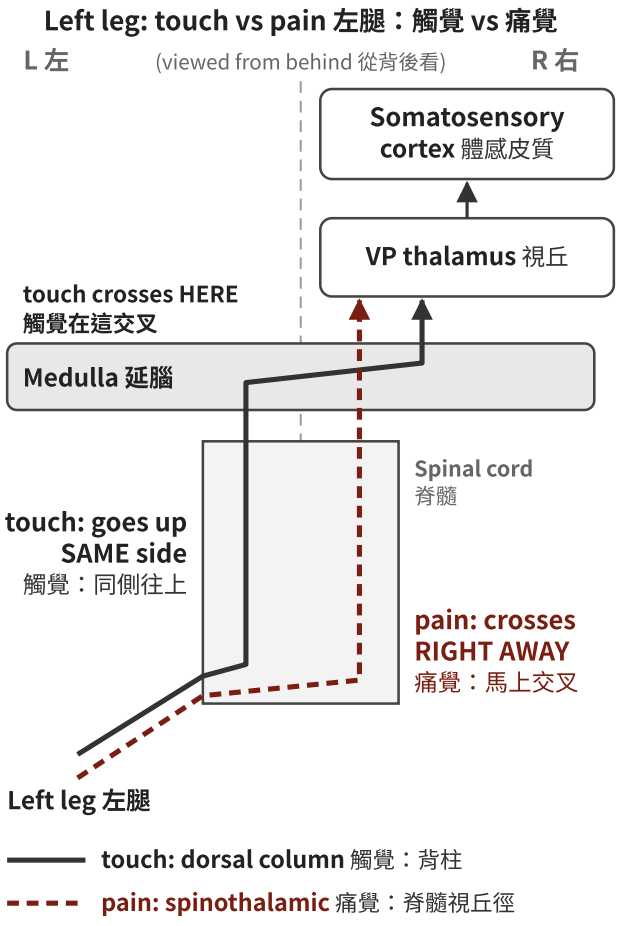

**Flag example: green, black, yellow flag → you see red, white, blue on the wall.**
綠黑黃國旗，看白牆變成紅白藍。

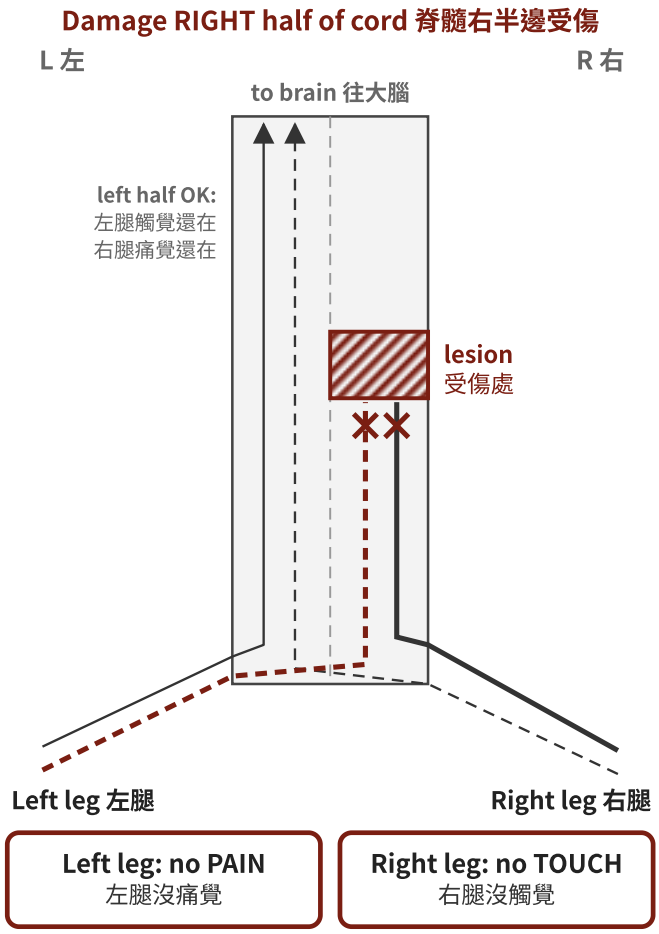
## 4. SG5｜聲音的路 Sound path （p.210 to 221）




**Ossicles (malleus, incus, stapes): air → fluid without losing energy (p.214 left col. lines 1 to 6).**
**Stapes pushes oval window → round window bulges out = escape valve (p.215 left col. lines 12 to 16).**
**First step is mechanical, not voltage: mechanically gated K+ channels (p.215 right col. lines 9 to 11).**
選項只差 K/Na 時選 K：內淋巴液鉀多鈉少（p.215 左欄 1 到 2 行）。教授 11/3 口頭說鈉。

## 5. SG8｜脊髓：觸覺 vs 痛覺 Touch vs pain pathways （p.228 to 231, 235 to 236）

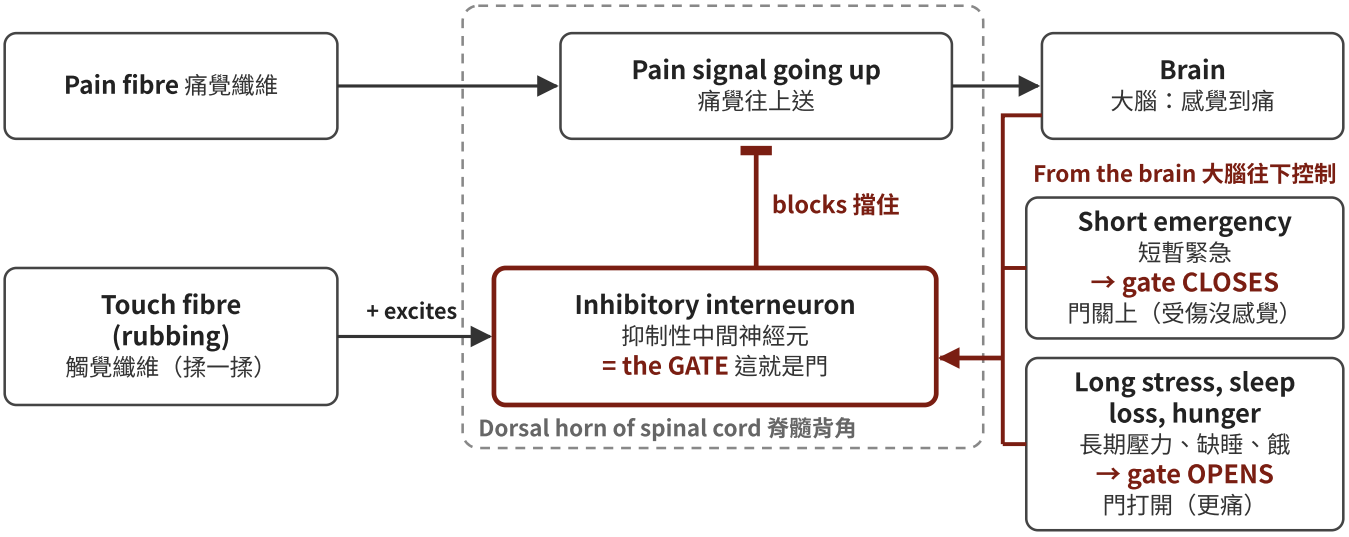



**傷哪邊，哪邊沒觸覺；痛覺在另一邊**

**Touch is lost on the SAME side as the damage; pain is lost on the OTHER side.**

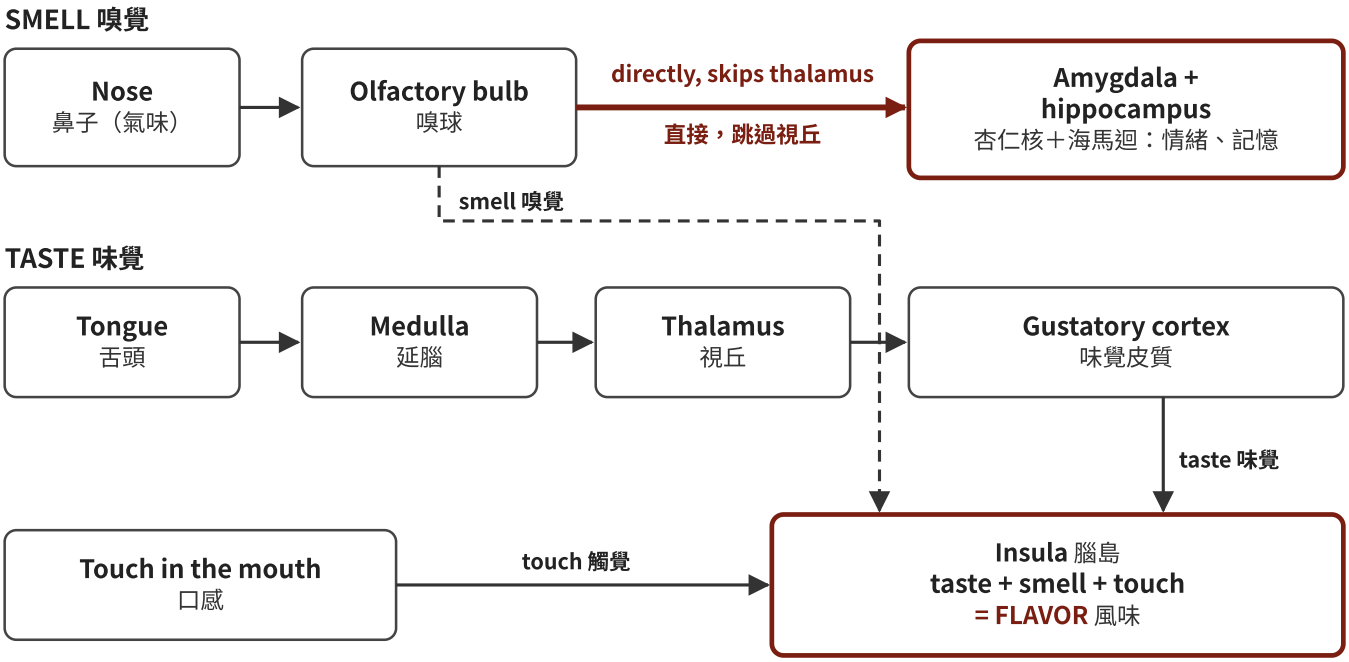
## 6. SG10｜閘門理論 Gate theory of pain （p.235 to 237）




**Touch closes the gate from BELOW; the brain controls it from ABOVE. Not distraction.**

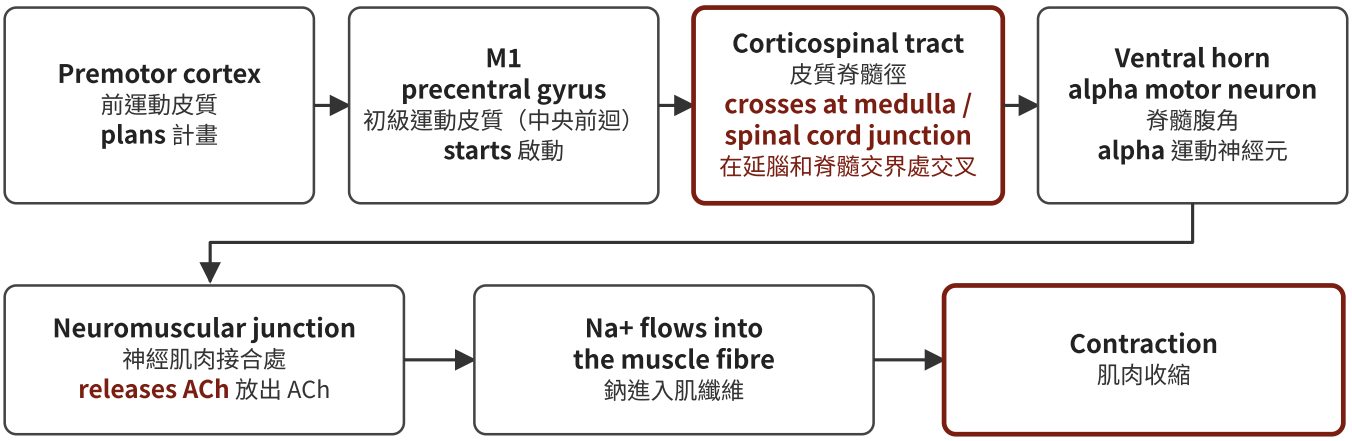
觸覺（揉一揉）從下面關門；大腦從上面控制。不是轉移注意力，痛覺訊號真的變少。

## 7. SG11｜嗅覺 vs 味覺 Smell vs taste （約 p.239 to 246）



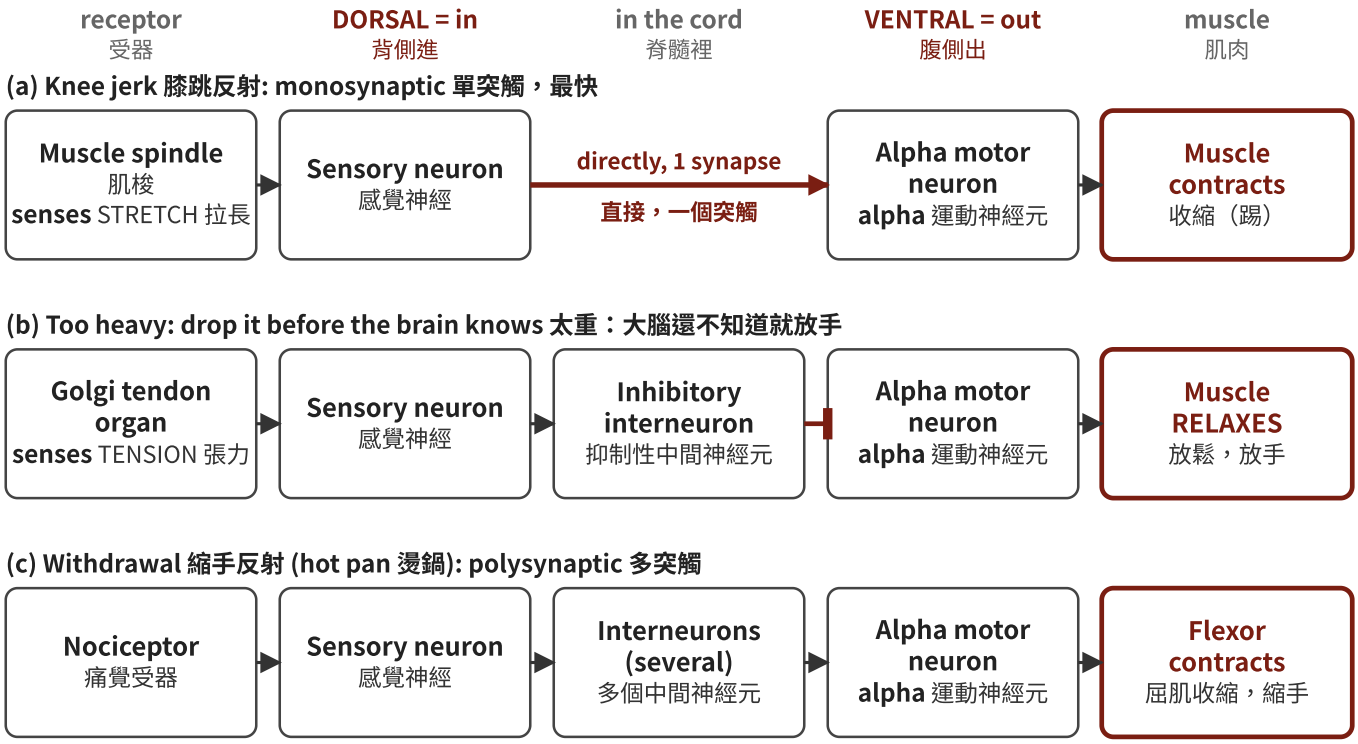

**Only smell skips the thalamus; other senses stop there first.** 只有嗅覺跳過視丘。
**Emotional value of flavor → orbitofrontal cortex.** 風味的情緒評價到眼眶額葉皮質。
**Module video says insula; textbook may say orbitofrontal.** 考場翻 p.239 to 246 確認。

## 8. SG13｜往下的運動路徑 Descending pathway （p.254 to 263）



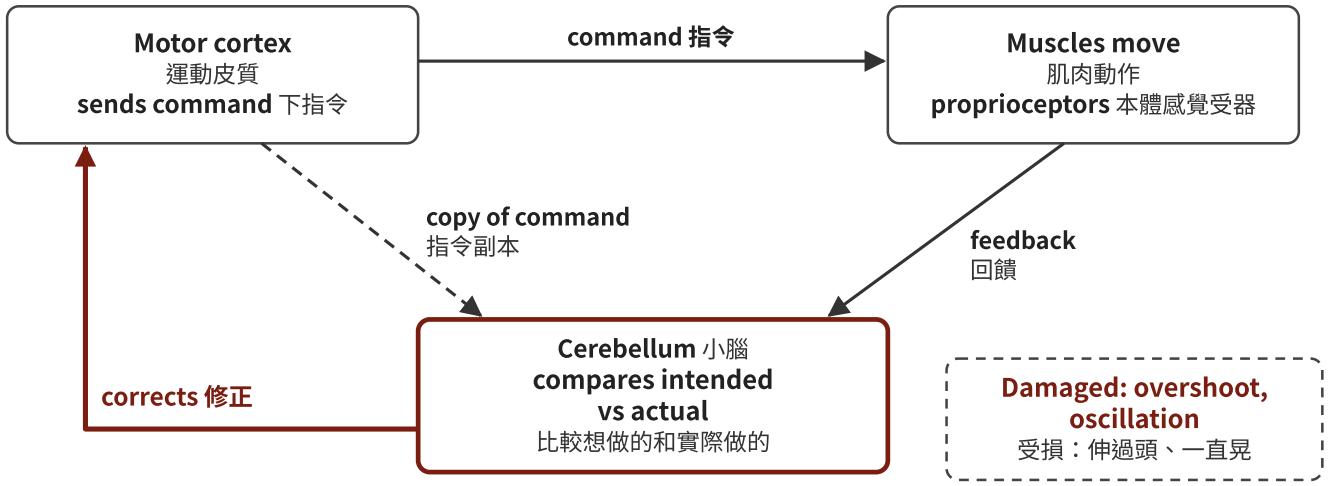

**Junction uses ACh, not dopamine.** 接合處是 ACh，不是多巴胺。
**Asked where it CROSSES: the junction. The ventral horn is the END point.** 問交叉答交界處；腹角是終點。
**Left hemisphere stroke → right hand weak.** 左腦中風 → 右手無力。

## 9. SG12 / SG13｜三種反射 Reflex arcs （p.255 to 258）




**Dorsal in, ventral out.** 背進腹出。
**Reciprocal inhibition: biceps contracts while triceps relaxes.** 互相抑制：二頭肌收縮時三頭肌放鬆。
**All done in the cord: a cut cord still kicks, but he feels nothing.** 脊髓斷了還會踢，但沒感覺。

## 10. SG15｜小腦修正迴路 Cerebellum error correction （p.263 to 264, 266）




**Only the cerebellum gets BOTH the command copy and the feedback.** 題目的 both 是關鍵。
**Basal ganglia = gas and brakes, not quality control.** 基底核是油門煞車，不是品管員。

## 11. SG16｜油門和煞車 Gas and brakes (basal ganglia) （p.264, 274 to 276, 279）

**GAS 油門 = substantia nigra, dopamine** 黑質多巴胺
**BRAKES 煞車 = striatum (caudate, putamen)** 紋狀體（尾核、殼核）
**Steering wheel 方向盤 = motor cortex** 運動皮質

### Sitting still 坐著不動

GAS ▬▬▬▬▬
BRAKES ▬▬▬▬▬

**balance: both at full power**
平衡：兩邊都全力

**not resting** 不是休息
**equilibrium** 力量相等

### Parkinson 帕金森

GAS ▬▬▭▭▭
BRAKES ▬▬▬▬▬

**gas too weak, brakes normal**
油門不夠，煞車正常

**rest tremor** 靜止會抖
**hard to start** 很難開始動
**depression first** 先憂鬱
**one side first** 先單側

### Huntington 杭丁頓

GAS ▬▬▬▬▬
BRAKES ▬▭▭▭▭

**brakes broken**
煞車壞了

**jerky movements** 停不下來
**dominant gene** 顯性基因
**caudate + putamen shrink**
尾核、殼核萎縮

Depression first: 教授版本（10/22），黑質的多巴胺也送到快樂中樞 nucleus accumbens。
課本解剖：送到依核的主要是 VTA。

## 12. SG18 to 20｜體溫 Temperature and the set point （Ch9 p.282 to 311）

**Set point** 設定點 37°C
**hypothalamus, preoptic area**
下視丘，視前區

**below 低於** → **Too cold** 太冷
**shiver** 發抖
**vasoconstrict** 血管收縮

**above 高於** → **Too hot** 太熱
**sweat** 流汗
**vasodilate** 血管擴張

### Fever 發燒

**Pyrogens**
致熱原（感冒）
→ **Raise the set point**
設定點被調高
→ **Normal temperature feels cold**
正常體溫被當成太冷
→ **Shivering under blankets**
發抖，蓋被子

**Homeostasis = the concept; set point = the setting (37°C).** 恆定是概念，設定點是具體設定。

## 13. SG21 / SG22 | 兩種口渴 Two thirsts and tonicity (Ch9 p.282 to 311)

**OSMOTIC thirst** 滲透壓口渴（吃鹹，沒失血）　**Fix: plain water** 補水就好

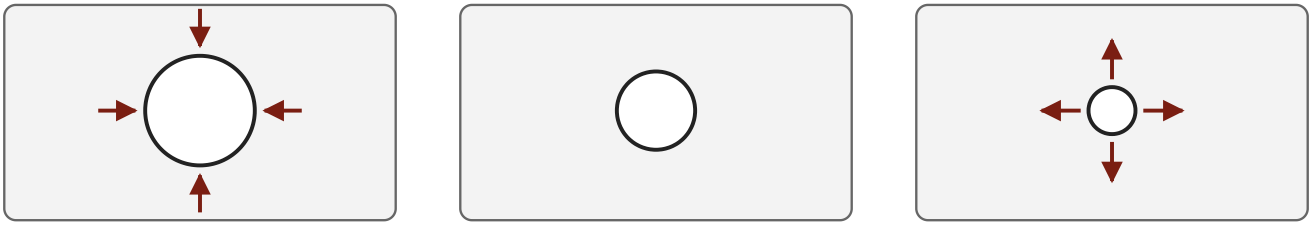
**Salty food** 吃鹹 → **Extracellular fluid hypertonic** 細胞外高張 → **Cells shrink** 細胞縮水 → **OVLT osmoreceptors** 滲透壓受器 → **Thirst + ADH** 口渴＋ADH

**HYPOVOLEMIC thirst** 低血容量口渴（失血、拉肚子）　**Needs water + salt** 要補水＋鹽

**Blood loss** 失血 → **Volume drops** 血量下降 → **Baroreceptors + kidneys** 壓力受器＋腎臟 → **Renin → angiotensin II** → **Subfornical organ** 穹窿下器官 → **Thirst + ADH** 口渴＋ADH

**Plain water alone dilutes the blood.** 只喝水會把血稀釋。　**Both thirsts feel the same.** 兩種口渴感覺一樣。

**A cell placed in a solution 細胞泡在溶液裡**



**Hypotonic** 低張
**outside dilute → water IN**
外面稀 → 水進去 → 脹大
例：很稀的點滴

**Isotonic** 等張
**same as inside**
一樣濃 → 不變

**Hypertonic** 高張
**outside salty → water OUT**
外面濃 → 水跑出來 → 縮水
例：3% 鹽水、吃鹹

## 14. SG23 / SG24 | 血糖和食慾 Glucose and feeding (Ch9 p.282 to 311)

**Blood glucose** 血糖（胰臟 pancreas）

**High blood glucose** 血糖高（吃完飯） → **Pancreas: INSULIN** 胰臟放胰島素 → **Glucose into cells, stored as glycogen** 葡萄糖進細胞、存成肝醣

**Low blood glucose** 血糖低（馬拉松沒吃） → **Pancreas: GLUCAGON** 胰臟放升糖素 → **Glycogen → glucose** 肝醣變回葡萄糖，大腦有糖

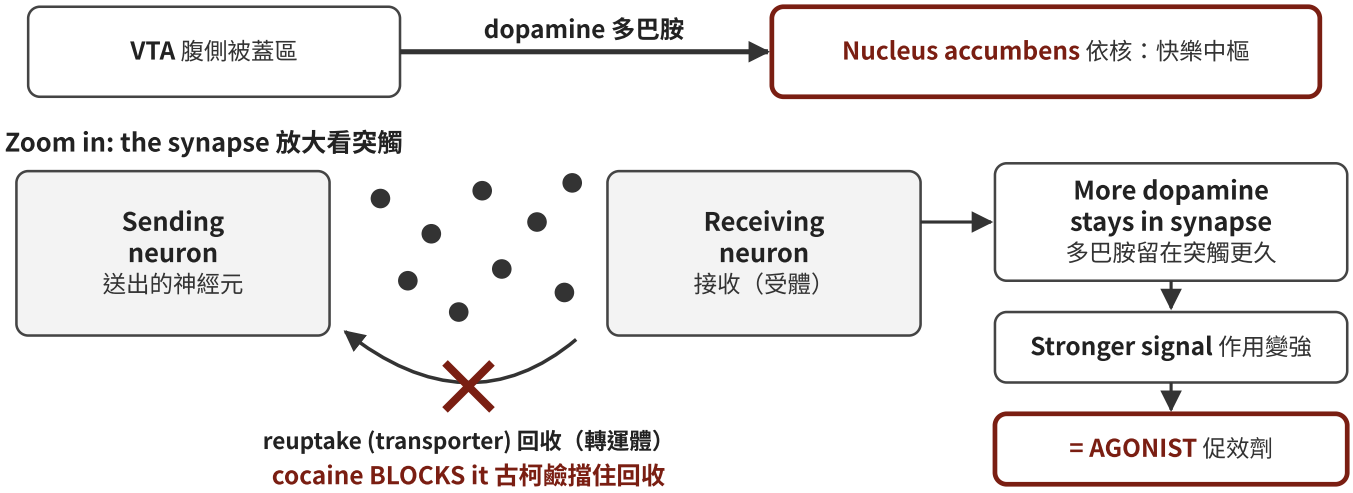
**Hunger and fullness** 餓和飽（下視丘 hypothalamus）

**Empty stomach** 空腹的胃 → **Ghrelin** 「我餓了」 → **Arcuate nucleus** 弓狀核 → **Orexin** 外側下視丘 → **EAT** 想吃

**Fat cells** 脂肪 → **Leptin** 「存夠了」 → **Arcuate nucleus** 弓狀核 → **STOP eating** 停，不吃

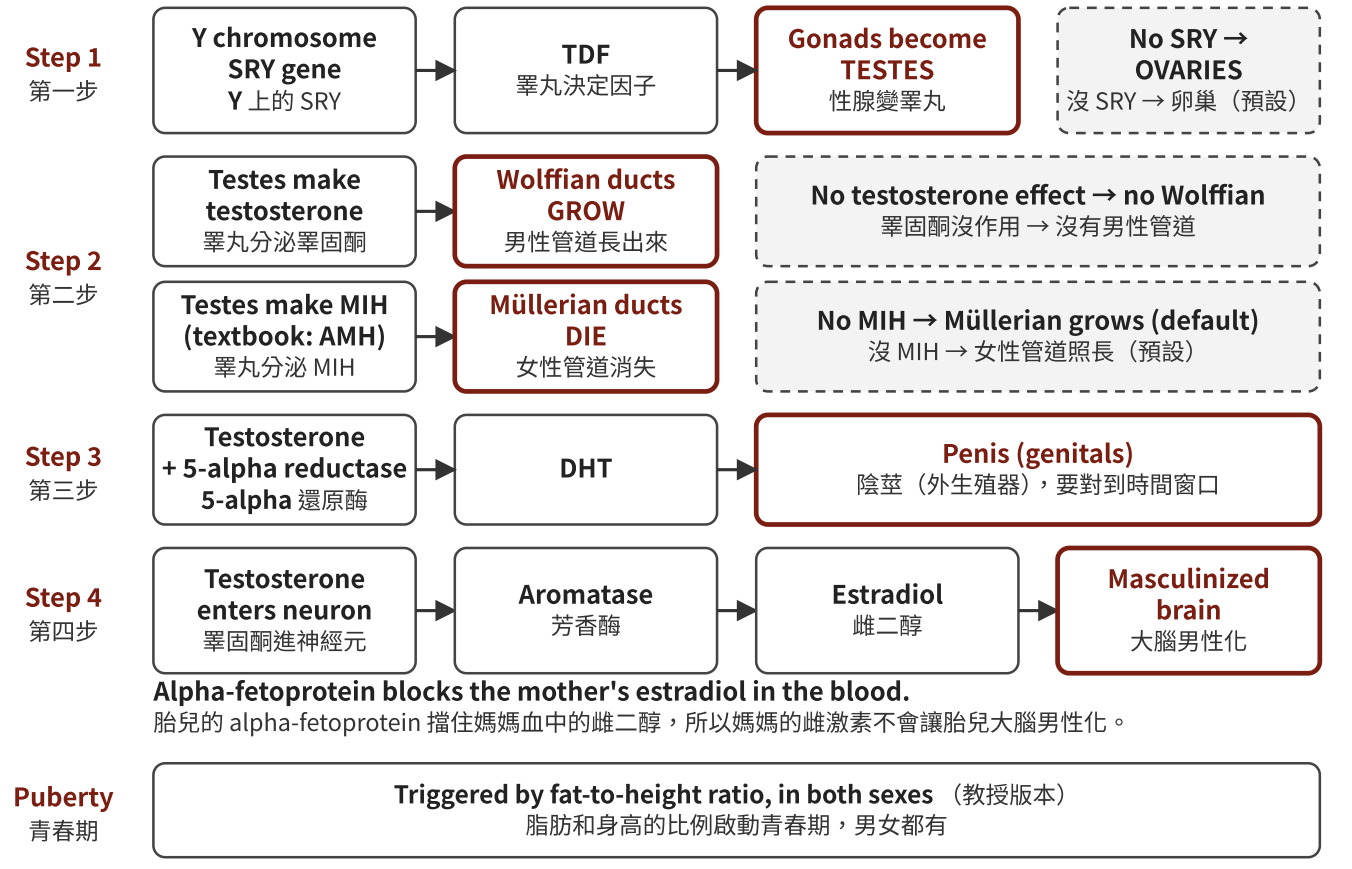
**Obesity: leptin is high but the brain stops listening (leptin resistance).** 胖的人 leptin 高還是餓：阻抗。
**Swap trap: ghrelin = stomach, hungry; leptin = fat, full.** 對調陷阱。

## 15. SG25｜獎賞迴路和古柯鹼 Reward and cocaine （Ch9 p.282 to 311）




**It blocks reuptake, not the receptor, so the effect gets stronger.** 擋的是回收，不是受體，所以作用變強。

## 16. SG26｜性分化四步驟 Four steps of sexual differentiation （p.314, 317 to 318, 320, 324）




**Alpha-fetoprotein blocks the mother's estradiol in the blood.**
胎兒的 alpha-fetoprotein 擋住媽媽血中的雌二醇，所以媽媽的雌激素不會讓胎兒大腦男性化。

**Puberty** 青春期 — **Triggered by fat-to-height ratio, in both sexes** （教授版本）
脂肪和身高的比例啟動青春期，男女都有

**先問有沒有睪丸（MIH），再問睪固酮有沒有作用（Wolffian）**

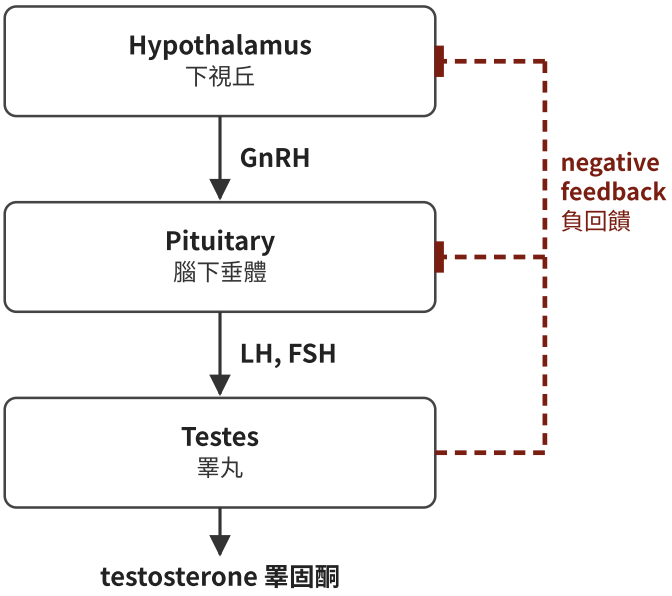
**First: are there testes? (MIH kills Müllerian). Then: does testosterone work? (Wolffian grows).**

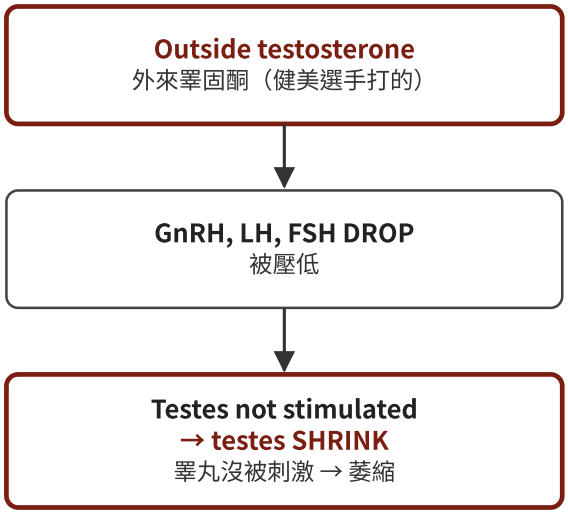
## 17. SG28｜發育異常對照表 Deviations （p.317 to 323）

| Condition<br>狀況 | Ducts<br>管道 | Genitals<br>外生殖器 | Brain<br>大腦 | Puberty / key<br>青春期／關鍵 |
|---|---|---|---|---|
| **XY, testosterone has no effect (AIS)**<br>受體沒反應 | **Neither**<br>兩套都沒有 | **Female**<br>女性 | **Female**<br>女性 | **Normal female**<br>正常女性發育<br>有胸部、不能生育 |
| **XY, no 5-alpha reductase**<br>缺 5-alpha 還原酶 | **Wolffian**<br>男性管道 | **Female-looking**<br>外觀女性 | **Masculinized**<br>男性化 | **Masculinizes at puberty**<br>青春期開始男性化 |
| **XY, mutated SRY**<br>**SRY** 不作用 | **Müllerian**<br>女性管道 | **Female**<br>女性 | **Female**<br>女性 | **SRY = first domino**<br>第一張骨牌 |
| **XX, excess androgens (CAH)**<br>**XX** 大量雄性素 | **Both**<br>兩套都有 | **Masculinized**<br>男性化 | **Masculinized**<br>男性化 | **No testes = no MIH**<br>沒睪丸＝沒 MIH（課堂版本） |
| **PMDS: MIH does not work**<br>**MIH** 沒作用 | **Both**<br>兩套都有 | **Male**<br>男性 | **Masculinized**<br>男性化 | **Testes yes, MIH broken**<br>有睪丸，MIH 壞了 |

**Rule: testes present → MIH → Müllerian dies; testosterone works → Wolffian grows.** 先問睪丸，再問睪固酮。

## 18. SG31｜荷爾蒙負回饋 Hormone feedback （p.337, 340, 342 to 343）




**Direction: DOWN, not up. 方向：壓低，不是調高。**

# 第六章 視覺 Vision

## 1. Sensation versus Perception （p.174）

**After a stroke, a woman's eyes test normal. She can describe a mug's color and edges but cannot say what it is. What is impaired?** [Sensation versus Perception]
中風後眼睛正常，說得出杯子的顏色和邊，但認不出是什麼。哪裡壞了？

→ **Perception is impaired; sensation is intact.**
Perception（知覺）。Sensation 正常。

**Key:** Can describe color and edges = the eyes transduce light, so sensation is intact; cannot name it = the brain's interpretation fails.
判斷關鍵：說得出顏色和邊＝眼睛有把光轉成訊號＝sensation 正常；認不出＝大腦詮釋壞了。

**Two friends see the same dress photo on the same screen; one sees blue-black, the other white-gold. Why?** [Sensation versus Perception]
同一張洋裝照片，一個看藍黑、一個看白金，為什麼？

→ **Sensation is the same, but perception differs: each brain assumes different lighting (color constancy).**
Sensation 一樣，perception 不同：大腦對光線的假設不同（color constancy）。

**Key:** The same light enters both people's eyes, so it is not a cone difference; the difference is how each brain interprets it.
判斷關鍵：光進眼睛一模一樣＝不是錐細胞的差別；差在大腦怎麼詮釋。

**A boy with no eyes navigates by tongue clicks; scans show echoes processed in visual cortex. What does this show?** [Sensation versus Perception]
沒有眼睛的男孩用彈舌回聲走路，回聲在視覺皮質處理。說明什麼？

→ **We see with the brain: the brain builds spatial perception from sound (Ben Underwood).**
We see with the brain：大腦用聲音建出空間知覺（Ben Underwood）。

**Key:** No eyes = no sensation of light, so the sense of space must come from the brain.
判斷關鍵：沒有眼睛＝沒有光的 sensation；空間感來自大腦。

## 2. Bottom-Up versus Top-Down （p.174, 196）

**An author proofreads her essay three times and misses 'the the'; a stranger spots it at once. Why?** [Bottom-Up versus Top-Down]
作者校稿三次沒看到 the the，陌生人一眼看到。為什麼？

→ **The author uses top-down processing: her expectation overrides what her eyes see. The stranger uses bottom-up processing.**
作者是 top-down：期待蓋過了眼睛看到的。陌生人是 bottom-up。

**Key:** Reading with knowledge and expectation = top-down; building up letter by letter = bottom-up, which is what catches the error.
判斷關鍵：用知識和期待去讀＝top-down；從字母一個個組起來＝bottom-up（這才抓得到錯）。

**You see a word for the first time in a language you don't know, letter by letter. Which processing?** [Bottom-Up versus Top-Down]
第一次看一個不認得的外文字，一個字母一個字母看。哪一種？

→ **Bottom-up processing.**
Bottom-up。

**Key:** A new, unexpected stimulus relies more on bottom-up processing. Both kinds run at the same time; one is just stronger.
判斷關鍵：新的、沒有預期的刺激，大腦用 bottom-up 比較多。兩種同時都在用，只是哪個比較強。

## 3. Photons, Light, Color, Refraction, Eye Structure （p.175–181, 198–203）

**A spotlight keeps the same wavelength but sends many more photons per second. What changes?** [Photons, Light, Color, Refraction, Eye Structure]
聚光燈波長不變，每秒光子變多。看起來有什麼改變？

→ **The color stays the same; it looks brighter.**
顏色不變，變亮。

**Key:** More photons = greater amplitude = brightness; same wavelength = same color (hue).
判斷關鍵：光子變多＝amplitude 變大＝亮度；波長沒變＝顏色沒變。

**A swimmer sees blurry underwater but clearly with goggles; her eyes are healthy. Why?** [Photons, Light, Color, Refraction, Eye Structure]
游泳時水裡看不清楚，戴蛙鏡就清楚，眼睛正常。為什麼？

→ **Refraction: in water, light barely bends at the cornea, so the focus misses the retina.**
Refraction：光在水裡經過角膜幾乎不彎，焦點對不到視網膜。

**Key:** The question is about a physical property of light = refraction; it has nothing to do with chemical changes in photoreceptors.
判斷關鍵：問的是光的物理性質＝折射；跟感光細胞的化學變化無關。

**Eye drops keep the dark center opening wide in bright light. Which structure do they act on?** [Photons, Light, Color, Refraction, Eye Structure]
眼藥水讓瞳孔在強光下還是很大。作用在哪個構造？

→ **The iris.**
Iris（虹膜）。

**Key:** The iris is the muscle (the curtain); the pupil is only an opening; the optic disc is only the back exit.
判斷關鍵：虹膜是窗簾（肌肉）；瞳孔只是洞；視神經盤只是後門。

**A 55-year-old reads distant signs but holds the menu at arm's length. Why?** [Photons, Light, Color, Refraction, Eye Structure]
55 歲看得清遠處，菜單要拿很遠才看得到。為什麼？

→ **The lens has stiffened and cannot thicken to focus on near objects.**
Lens（水晶體）變硬，不能變厚對焦近物。

**Key:** The lens is the part that adjusts; the cornea's shape is fixed.
判斷關鍵：可以調整的是水晶體；角膜形狀固定。

## 4. Neurons and steps in visual processing, receptive fields, visual cortex （p.180–194, 205–206）

**A retina sits in total darkness. What are the photoreceptors doing?** [Neurons and steps in visual processing, receptive fields, visual cortex]
視網膜在全黑中，感光細胞在做什麼？

→ **Photoreceptors are active: cGMP is high, sodium channels are open, they are depolarized, and they release glutamate constantly, inhibiting bipolar cells.**
活躍：cGMP 高、鈉通道開、去極化、一直放 glutamate，壓住雙極細胞。

**Key:** Professor's version: the vampire rules in the dark. "Light activates photoreceptors" is wrong.
判斷關鍵：教授版本：吸血鬼在黑暗中統治。「光活化感光細胞」是錯的。

**A drug stops cGMP from being broken down. The person is in a bright room. What happens to bipolar cells?** [Neurons and steps in visual processing, receptive fields, visual cortex]
藥物讓 cGMP 不被分解，人在很亮的房間。雙極細胞怎樣？

→ **Bipolar cells stay inhibited, so the person cannot see.**
一直被壓住，看不見。

**Key:** cGMP not broken down = photoreceptors act as if still in the dark = they keep releasing glutamate.
判斷關鍵：cGMP 不分解＝感光細胞以為還在黑暗＝繼續放 glutamate。

**A faint star disappears when looked at directly but appears when looking to the side. Why?** [Neurons and steps in visual processing, receptive fields, visual cortex]
暗的星星正看看不到，往旁邊看才看到。為什麼？

→ **The periphery has rods: they are very sensitive, and many rods converge on one bipolar cell.**
周邊是 rods：敏感、多個 rods 匯聚到一個雙極細胞。

**Key:** The center has cones, which need bright light; peripheral rods are sensitive but blurry and see no color.
判斷關鍵：中央是 cones，要亮光；周邊 rods 敏感但模糊、沒顏色。

**Stare at a green-black-yellow flag, then a white wall: you see red-white-blue. Where and why?** [Neurons and steps in visual processing, receptive fields, visual cortex]
盯著綠黑黃國旗再看白牆，看到紅白藍。在哪一層、為什麼？

→ **Opponent-process in ganglion cells: the side that was suppressed rebounds after you look away.**
神經節細胞的 opponent-process：一直被壓住的那一邊，移開視線後反彈。

**Key:** Green pairs with red, yellow with blue, black with white. Trichromatic theory (cones) is still correct; the two theories do not conflict.
判斷關鍵：綠↔紅、黃↔藍、黑↔白。三色說（錐細胞）照樣是對的，兩者不矛盾。

**An on-center cell fires to a small spot on its center, but only weakly to light over its whole field. Why?** [Neurons and steps in visual processing, receptive fields, visual cortex]
on-center 細胞照中心反應很強，整片照滿反應很弱。為什麼？

→ **The inhibitory surround cancels the excitatory center (lateral inhibition, horizontal cells).**
外圈抑制抵消了中心的興奮（lateral inhibition，水平細胞）。

**Key:** Light on the center = excitation; light on the surround = inhibition; whole field = they cancel, so the brain attends most to edges.
判斷關鍵：照中心＝興奮；照外圈＝抑制；整片＝抵消。所以大腦最注意邊緣。

**After a stroke a man recognizes faces and objects, but pouring coffee looks like frozen snapshots. Damage?** [Neurons and steps in visual processing, receptive fields, visual cortex]
中風後認得臉和東西，倒咖啡卻像一格一格的照片。哪裡壞了？

→ **Damage to area MT in the dorsal stream (akinetopsia).**
Dorsal stream 的 MT（akinetopsia）。

**Key:** Recognizes faces and objects = the ventral (what) stream is intact, so cross out prosopagnosia first. Objects OK but not faces = prosopagnosia.
判斷關鍵：題目說認得出臉和東西＝ventral（what）正常，先劃掉臉盲。只認得出東西、認不出臉＝正是臉盲。

**Neuron X fires to a vertical bar at one exact spot; neuron Y fires to it anywhere, best when moving. Which is which?** [Neurons and steps in visual processing, receptive fields, visual cortex]
X 只對固定位置的直線放電；Y 哪裡都行、會動最強。

→ **X is a simple cell; Y is a complex cell.**
X＝simple cell，Y＝complex cell。

**Key:** Fixed position = simple cell; any position and prefers motion = complex cell.
判斷關鍵：固定位置＝simple；位置不限、喜歡動＝complex。

**Why can't you see the blood vessels lying in front of your retina?** [Neurons and steps in visual processing, receptive fields, visual cortex]
為什麼看不到視網膜前面的血管？

→ **The stimulus is constant, so the brain filters it out and fills it in.**
刺激是恆定的，大腦把它過濾掉、補起來。

**Key:** Professor 11/3: constant things are not seen. Signals go one way only (photoreceptor → bipolar cell → ganglion cell), and the ganglion cell is the last layer.
判斷關鍵：教授 11/3：constant 的東西看不到。訊息單向：感光細胞 → 雙極細胞 → 神經節細胞，不能倒傳；節細胞是最後一層。

# 第七章 聽覺、體感、嗅味覺（p.209–246）

## 5. Sound waves, ear structure, auditory processing, auditory cortex （p.210–221）

**A synthesizer's wave gets taller at the same rate, then repeats faster at the same height. What do listeners perceive?** [Sound waves, ear structure, auditory processing, auditory cortex]
聲波先變高、頻率不變，再變快、高度不變。聽起來怎樣？

→ **First it sounds louder, then it sounds higher in pitch.**
先變大聲，再變高音。

**Key:** Wave height = amplitude = loudness; cycles per second = frequency = pitch.
判斷關鍵：高度＝amplitude＝響度；每秒週期數＝frequency＝音高。

**After years near loud machines, an engineer can't hear high pitches but hears bass. Where is the damage?** [Sound waves, ear structure, auditory processing, auditory cortex]
長年噪音，聽不到高音，低音還聽得到。哪裡壞了？

→ **The base of the basilar membrane (near the oval window).**
基底膜底部 base（靠近 oval window）。

**Key:** High pitch is at the base, low pitch at the apex. Professor's ribbon analogy: small waves only move the start of the ribbon.
判斷關鍵：高音在底部、低音在頂端。教授緞帶比喻：小波浪只動到開頭。

**What path do the remaining sounds take to conscious hearing?** [Sound waves, ear structure, auditory processing, auditory cortex]
剩下的聲音怎麼傳到意識？

→ **Auditory nerve → MGN (thalamus) → auditory cortex in the temporal lobe.**
Auditory nerve → MGN（視丘）→ 顳葉聽覺皮質。

**Key:** MGN = hearing; LGN is vision and VP is touch, so do not mix them up.
判斷關鍵：LGN 是視覺、VP 是觸覺，不要混。

**Hair cell stereocilia bend toward the tallest one. What happens first?** [Sound waves, ear structure, auditory processing, auditory cortex]
毛細胞的毛往最高那根彎，第一件發生的是什麼？

→ **Channels are pulled open mechanically, positive ions flow in, and the cell depolarizes.**
通道被機械性拉開，正離子流進去，去極化。

**Key:** The first step is mechanical, not voltage (professor 11/3). Textbook p.215: mechanically gated $K^+$ channels open, endolymph $K^+$ flows in; the 11/3 Kahoot said sodium. If options differ only in $K^+$ vs $Na^+$, choose $K^+$.
判斷關鍵：一開始不是電壓（教授 11/3）。課本 p.215 右欄 9 到 11、25 到 26 行：鉀流進去；左欄 1 到 2 行：內淋巴液鉀多鈉少。教授口頭說鈉。只差 K/Na 選 K。

## 6. McGurk Effect, Associative Cortices （p.219, 221）

**Audio says 'ba', lips say 'ga', viewers hear 'da'; with eyes closed they hear 'ba'. What does this show?** [McGurk Effect, Associative Cortices]
聲音 ba、嘴型 ga，聽到 da；閉眼聽到 ba。說明什麼？

→ **Association cortex integrates vision and hearing; vision rewrites perception.**
聯合皮質整合視覺和聽覺，視覺改寫了知覺。

**Key:** Eyes closed = "ba" again, so the ears did not change; the change is in the brain. You hear with your brain.
判斷關鍵：閉眼又聽到 ba＝耳朵沒變；改變在大腦。You hear with your brain。

**Stopped at a red light, the car beside you rolls back and you feel yourself rolling forward. Why?** [McGurk Effect, Associative Cortices]
停紅燈旁邊車往後退，覺得自己往前滑。為什麼？

→ **When the senses conflict, vision wins and overrides the vestibular sense.**
感覺衝突時視覺贏，蓋過前庭覺。

**Key:** "Although" the inner ear fluid did not move = the answer must explain this conflict. Neuroplasticity does not happen in seconds.
判斷關鍵：題目寫 although 內耳液體沒動＝答案要解釋這個矛盾。神經可塑性不會在幾秒內發生。

**A hidden real hand and a visible rubber hand are stroked together; a hammer hits the rubber hand and she flinches. Why?** [McGurk Effect, Associative Cortices]
真手藏起來和假手一起被摸，鎚子打假手她嚇到。為什麼？

→ **Vision overrides proprioception, and the brain adds the rubber hand to its body map.**
視覺蓋過本體感覺，大腦把假手收進身體地圖。

**Key:** Same rule as the McGurk effect, the red light illusion, and the mirror box: vision wins.
判斷關鍵：跟 McGurk、紅燈、鏡箱同一條規則。

## 7. Somatosensory cortex, motor cortex, receptor types （p.227–232, 235, 265）

**A patient can't feel deep pressure or vibration on her fingertips but a hot mug still hurts. Why?** [Somatosensory cortex, motor cortex, receptor types]
指尖感覺不到壓力和震動，摸熱杯子卻會痛。為什麼？

→ **The Pacinian corpuscles are damaged, while the separate pain receptors still work.**
Pacinian 小體壞了，獨立的痛覺受器還正常。

**Key:** "Whereas" is not a trap: split the option in two, and both halves must be true. Touch and pain use two different sets of receptors.
判斷關鍵：whereas 不是陷阱：切兩半，兩半都要對。觸覺和痛覺是兩組受器。

**A boy has burns and fractures he never noticed; touch, reflexes, axons and myelin are normal. Cause?** [Somatosensory cortex, motor cortex, receptor types]
男孩受傷都沒感覺，觸覺、反射、軸突、髓鞘都正常。原因？

→ **His nociceptors do not function.**
Nociceptors 不作用。

**Key:** Everything else is normal, so only the receptor at the very front is left. Pain's job is to warn.
判斷關鍵：前面全部正常＝只剩最前端的受器。痛的功能是警告。

**Sudden pressure on a Pacinian corpuscle fires the fiber; steady pressure soon stops it. Why?** [Somatosensory cortex, motor cortex, receptor types]
突然壓 Pacinian 小體會放電，持續壓很快就停。為什麼？

→ **Deformation pulls channels open mechanically and sodium enters; steady pressure is ignored (rapid adaptation).**
變形把通道機械性拉開、鈉進去；持續壓力它不理（快速適應）。

**Key:** Skin has no hair cells, and touch receptors are not opened by a neurotransmitter.
判斷關鍵：皮膚沒有毛細胞；觸覺受器不靠神經傳導物質打開。

**During surgery, stimulating one strip makes the right thumb tingle; the strip in front makes it twitch. Where?** [Somatosensory cortex, motor cortex, receptor types]
刺激一條皮質右拇指刺刺的，前面那條讓它抽動。在哪？

→ **Both are in the left hemisphere: tingling = postcentral gyrus (somatosensory); twitching = precentral gyrus (motor).**
都在左半腦：刺刺＝postcentral（體感），抽動＝precentral（運動）。

**Key:** Front = motor, back = sensory, and both control the opposite side. The two maps run in parallel, but the sensory map is set by skin input.
判斷關鍵：前面是運動，後面是感覺；都管對側。兩張地圖平行，但感覺地圖由皮膚輸入決定。

## 8. Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways (p.228–231, 235–236)

**Damage to the RIGHT half of the spinal cord at the mid-back; he can still bend his left knee. Sensory loss?** [Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways]
脊髓右半邊受傷，左膝還能彎。感覺怎麼失去？

→ **Loss of touch in the right leg; loss of pain in the left leg.**
右腿沒觸覺，左腿沒痛覺。

**Key:** Touch is lost on the injured side and pain on the opposite side. "Can bend the left knee" is a distractor, so draw it first.
判斷關鍵：傷哪邊，哪邊沒觸覺；痛覺在另一邊。「左膝能彎」是干擾。先畫圖。

**Damage to the LEFT half of the thoracic cord; axons and myelin are normal. Loss below the lesion?** [Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways]
胸椎左半邊受傷，軸突髓鞘正常。損傷以下怎樣？

→ **Loss of touch on the left side; loss of pain on the right side.**
左邊沒觸覺，右邊沒痛覺。

**Key:** "Axons and myelin are normal" is a borrowed distractor and has nothing to do with the answer.
判斷關鍵：axons and myelin normal 是借來的干擾，跟答案無關。

**Injury at T10: no sensation below the navel, but arms and chest are fine. Principle?** [Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways]
T10 受傷，肚臍以下沒感覺，手臂胸部正常。原理？

→ **Dermatomes: each area of skin maps to one level of the spinal cord, so signals from below the injury cannot go up.**
Dermatomes：每塊皮膚對應脊髓一個高度，損傷以下的上不去。

**Key:** Option B's weak point is "immediate": plasticity does not happen right away.
判斷關鍵：B 的破綻是 immediate：可塑性不會立刻發生。

**Two caliper points feel like two on the fingertip but one on the upper arm. Why?** [Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways]
兩點在指尖是兩點，在上臂是一點。為什麼？

→ **Fingertips have dense receptors with small receptive fields; the upper arm has sparse receptors with large receptive fields.**
指尖受器密、感受野小；上臂受器疏、感受野大。

**Key:** Two points = dense receptors, small fields. Remember: where a pathway crosses depends on the type of sense (touch vs pain), not the body part.
判斷關鍵：交叉點看感覺種類（觸覺 vs 痛覺），不看身體部位。

**After spinning and stopping, you still feel dizzy. Why?** [Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways]
轉圈停下來還是頭暈。為什麼？

→ **Fluid in the semicircular canals keeps flowing by inertia and keeps pushing the hair cells.**
半規管裡的液體有慣性，還在流動、推動毛細胞。

**Key:** Still dizzy after stopping = the fluid is still moving. Vestibular hair cells are mechanoreceptors too.
判斷關鍵：前庭毛細胞也是機械受器。

## 9. Homunculus, plasticity, phantom limbs and treatment （p.231–233）

**A violinist's left-hand finger area in somatosensory cortex is enlarged. Why?** [Homunculus, plasticity, phantom limbs and treatment]
小提琴家左手手指的體感皮質區變大。為什麼？

→ **Long-term use causes neuroplasticity: the area borrows neighboring neurons.**
長期使用 → 神經可塑性，借用鄰近神經元。

**Key:** The change is in the brain (central), not more receptors in the skin. Left hand → right hemisphere.
判斷關鍵：改變在大腦（中樞），不是皮膚受器變多。左手 → 右半腦。

**Weeks after losing his right hand, stroking his right cheek feels like touch on the missing fingers. Why?** [Homunculus, plasticity, phantom limbs and treatment]
右手截肢幾週後，摸右臉覺得手指被碰。為什麼？

→ **On the homunculus the face lies next to the hand; the left hemisphere's hand area gets no input and is taken over by face signals.**
小人圖上臉在手旁邊；左半腦的手區沒有輸入，被臉的訊號佔走。

**Key:** Right hand → left hemisphere (contralateral).
判斷關鍵：右手 → 左半腦（對側）。

**An amputee's phantom hand is clenched; seeing the intact hand open in a mirror box relieves it. Why?** [Homunculus, plasticity, phantom limbs and treatment]
幻肢一直握拳，鏡箱看到手張開就緩解。為什麼？

→ **Vision supplies the missing feedback, so the motor cortex stops sending the clench command.**
視覺補上缺少的回饋，運動皮質停止一直下令握拳。

**Key:** No feedback = a positive feedback loop that clenches tighter and tighter. The missing limb has no tendon, so it cannot activate the Golgi tendon organ.
判斷關鍵：沒有回饋＝正回饋迴圈越握越緊。斷肢沒有肌腱，不會啟動 Golgi tendon organ。

## 10. Gate theory of pain （p.235–237）

**After bumping an elbow, rubbing it lessens the pain. How?** [Gate theory of pain]
撞到手肘揉一揉比較不痛。怎麼做到的？

→ **Touch signals activate inhibitory interneurons in the spinal cord, which block ascending pain (the gate closes).**
觸覺訊號啟動脊髓的抑制性中間神經元，擋住往上的痛覺（門關上）。

**Key:** It is not distraction; less pain signal actually reaches the brain. Rubbing does not destroy pain nerve endings.
判斷關鍵：不是轉移注意力，是真的比較少痛覺到大腦。揉一揉不會破壞痛覺神經末梢。

**A player doesn't notice a deep cut during a game; a week of finals stress makes a paper cut hurt more. Why?** [Gate theory of pain]
比賽中受傷沒感覺，期末壓力一週後紙割傷特別痛。為什麼？

→ **A short-term emergency closes the gate; long-term stress opens the gate.**
短暫緊急把門關上；長期壓力把門打開。

**Key:** Do not reverse the direction: emergency = gate closed, chronic stress = gate open.
判斷關鍵：方向不要對調。

**Pain is worse on days of sleep loss and stress; nerves and healing are normal. Mechanism?** [Gate theory of pain]
缺睡、壓力大的日子更痛，神經和傷口都正常。機制？

→ **Descending inhibitory pathways weaken, so more pain signals reach the brain.**
往下的抑制路徑變弱，更多痛覺訊號上到大腦。

**Key:** The question says peripheral nerves are normal = cross out peripheral options first. Receptors do not change by themselves.
判斷關鍵：題目說周邊神經正常＝先劃掉周邊選項。受器不會自己變身。

## 11. Smell and taste pathways （約 p.239–246）

**Smelling lavender brings a wave of childhood emotion before she can name the scent. Why so fast?** [Smell and taste pathways]
聞到薰衣草，還沒認出就湧上童年情緒。為什麼這麼快？

→ **Smell bypasses the thalamus and goes directly to the amygdala and hippocampus.**
嗅覺跳過視丘，直通杏仁核和海馬迴。

**Key:** Senses in the head do not use the spinal cord. Every other sense relays through the thalamus first.
判斷關鍵：頭上的感覺都不走脊髓。其他感覺都要先到視丘轉接。

**With a pinched nose, she tastes sweetness but can't tell apple from pear. Where is flavor built?** [Smell and taste pathways]
捏著鼻子只嚐到甜，分不出蘋果和梨。風味在哪合成？

→ **The insula, where taste, smell, and touch meet; emotional value goes to the orbitofrontal cortex. (Module video version; the textbook may say orbitofrontal, so check p.239 to 246.)**
Insula（腦島），味覺、嗅覺、觸覺會合；情緒評價到 orbitofrontal cortex。（Module 影片版本；課本可能寫 orbitofrontal，考場翻 p.239 到 246 確認）

**Key:** The tongue tastes only basic tastes; flavor = taste + smell + mouthfeel.
判斷關鍵：舌頭只嚐基本味；風味＝味覺＋嗅覺＋口感。

# 第八章 動作系統（p.248–279）

## 12. Ascending pathway from proprioceptors （p.255–258）

**A patient moves his legs on command but stumbles in the dark and can't tell where his feet are. What is lost?** [Ascending pathway from proprioceptors]
叫他動腳動得了，暗處卻不知道腳在哪。失去什麼？

→ **Ascending proprioception is lost (muscle spindles and Golgi tendon organs to the cerebellum and cortex).**
往上的本體感覺（肌梭、Golgi tendon organ 到小腦和皮質）。

**Key:** Can move on command = the descending pathway is intact; cannot tell where the feet are = the ascending signal is lost. (p.255 to 258)
判斷關鍵：能照指令動＝往下的路正常。（p.255–258）

## 13. Descending pathway, motor neurons, acetylcholine, reflexes （p.254–263）

**The motor cortex and alpha motor neurons fire normally, yet muscles barely contract. Which step fails?** [Descending pathway, motor neurons, acetylcholine, reflexes]
皮質和 alpha 運動神經元都正常，肌肉卻幾乎不收縮。哪一步壞？

→ **ACh cannot bind to the muscle's receptors, so the sodium channels do not open.**
ACh 接不上肌肉的受器，鈉通道打不開。

**Key:** The neuromuscular junction uses ACh, not dopamine. (p.254)
判斷關鍵：神經肌肉接合處是 ACh，不是 dopamine。（p.254）

**A left-hemisphere stroke weakens the right hand. Where did those axons cross?** [Descending pathway, motor neurons, acetylcholine, reflexes]
左半腦中風右手無力，軸突在哪交叉？

→ **At the junction of the medulla and spinal cord (pyramidal decussation).**
延腦和脊髓交界處（pyramidal decussation）。

**Key:** Asked where they cross = the junction; the ventral horn is the end point; the midbrain is the rubrospinal tract. (p.263)
判斷關鍵：問 cross 答交界處；腹角是終點；midbrain 是 rubrospinal。（p.263）

**A climber suddenly lets go of a ledge before deciding to, when forearm strain is extreme. Mechanism?** [Descending pathway, motor neurons, acetylcholine, reflexes]
攀岩者手臂張力太大，還沒決定就放手。機制？

→ **Golgi tendon organ → inhibitory interneuron → inhibits alpha motor neurons.**
Golgi tendon organ → 抑制性中間神經元 → 抑制 alpha 運動神經元。

**Key:** Tension = Golgi tendon organ; stretch = muscle spindle, which makes the muscle contract. (p.258)
判斷關鍵：張力＝tendon organ；拉長＝spindle（讓肌肉收縮）。（p.258）

**Complete mid-back cord injury: the knee kicks to a tap but he feels nothing. Why?** [Descending pathway, motor neurons, acetylcholine, reflexes]
胸椎完全斷，敲膝蓋會踢但沒感覺。為什麼？

→ **The knee jerk is monosynaptic and happens entirely in the spinal cord; sensation cannot travel up to the brain.**
膝跳是單突觸，整個在脊髓完成；感覺上不去大腦。

**Key:** "Is the reflex there?" and "Can he feel it?" have opposite answers. (p.256)
判斷關鍵：反射有沒有、感覺得到嗎，答案相反。（p.256）

**Touching a hot pan jerks the arm back; a tendon tap kicks the leg. How do the circuits differ?** [Descending pathway, motor neurons, acetylcholine, reflexes]
碰燙鍋縮手 vs 敲膝蓋踢腳，迴路差在哪？

→ **Withdrawal = nociceptors, polysynaptic; knee jerk = muscle spindle, monosynaptic.**
縮手＝痛覺受器、多突觸；膝跳＝肌梭、單突觸。

**Key:** Hot pan = pain receptor plus interneurons; tendon tap = stretch receptor, one synapse. Option D's trap swaps the receptors.
判斷關鍵：D 的陷阱是感受器對調。

**The biceps contracts while the triceps relaxes. What spinal mechanism?** [Descending pathway, motor neurons, acetylcholine, reflexes]
二頭肌收縮時三頭肌放鬆，什麼機制？

→ **Reciprocal inhibition.**
Reciprocal inhibition。

**Key:** Muscles can only pull, not push; if both sides pulled at once, the joint would lock. (p.259)
判斷關鍵：肌肉只會拉不會推，兩邊同時拉就卡住。(p.259)

**Fine eye muscles vs the calf: how do motor units differ?** [Descending pathway, motor neurons, acetylcholine, reflexes]
眼睛肌肉 vs 小腿肌肉，運動單位差在哪？

→ **Fine control = small motor units (one neuron controls few muscle fibers); force = large motor units.**
精細＝小單位（一個神經元管很少肌纖維）；出力＝大單位。

**Key:** Precise = small units, powerful = large units. Exact numbers are not tested. (p.254)
判斷關鍵：不考數字。(p.254)

## 14. Somatotopic map, cortically controlled FES, treating paralysis （p.265–266）

**Stimulating a small strip moves fingers and lips; an equal strip moves the whole leg. Principle?** [Somatotopic map, cortically controlled FES, treating paralysis]
一小塊皮質動手指和嘴唇，同樣大小動整條腿。原理？

→ **Somatotopic map: body parts with finer control get more cortex.**
Somatotopic map：越精細的部位分到越多皮質。

**Key:** Cortex area follows precision of control, not body size.
判斷關鍵：面積看精細度，不看身體大小。

**A man with a neck cord injury picks up a cup using a motor-cortex implant that stimulates forearm muscles. What is true?** [Somatotopic map, cortically controlled FES, treating paralysis]
頸椎受傷的人用運動皮質晶片刺激前臂拿杯子。哪個對？

→ **FES reads out the signal and bypasses the damaged cord, restoring some movement; it does not repair the injury.**
FES：讀出訊號、繞過受損脊髓，恢復部分動作，不修復損傷。

**Key:** FES reads signals OUT; DBS sends stimulation IN to the brain.
判斷關鍵：FES 讀出 OUT；DBS 送進 IN。

## 15. Cerebellum in error correction （p.263–264, 266）

**Reaching for a cup, his hand oscillates and overshoots repeatedly. Which structure gets both command copies and feedback?** [Cerebellum in error correction]
伸手拿杯子一直晃、伸過頭。哪個構造同時收指令副本和回饋？

→ **The cerebellum.**
Cerebellum。

**Key:** "Both" is the key word, since the somatosensory cortex receives only feedback. Overshoot = think cerebellum first. (p.263 to 264)
判斷關鍵：題目的 both 是關鍵；體感皮質只收回饋。看到 overshoot 先想小腦。（p.263–264）

**With prism glasses, a healthy person adapts after a few throws; a patient keeps missing. Damage?** [Cerebellum in error correction]
戴稜鏡丟飛鏢，正常人幾次就修正，病人一直偏。哪裡壞？

→ **The cerebellum (it corrects errors trial by trial).**
Cerebellum（一次一次修正錯誤）。

**Key:** Adapts after a few throws = error correction = cerebellum. The basal ganglia are the gas and brakes, not the quality checker.
判斷關鍵：基底核是油門煞車，不是品管員。

## 16. Motor disorders (Huntington and Parkinson) （p.264, 274–276, 279）

**A 62-year-old's hand trembles at rest and he freezes in doorways. Gas and brakes?** [Motor disorders (Huntington and Parkinson)]
62 歲靜止時手抖、走到門口卡住。油門煞車怎樣？

→ **Not enough gas (low dopamine from the substantia nigra), brakes normal: Parkinson's disease.**
油門不夠（黑質多巴胺少），煞車正常 → Parkinson。

**Key:** Tremor at rest = brakes overpower the gas; hard to start moving = brakes released but still no gas.
判斷關鍵：坐著抖＝煞車蓋過油門；要動更難＝放開煞車還是沒油。

**A woman in her forties with jerky movements she can't stop; her caudate and putamen have shrunk. Cause?** [Motor disorders (Huntington and Parkinson)]
四十幾歲抽動停不下來，尾核和殼核萎縮。原因？

→ **Degeneration of the striatum (the brakes): Huntington's disease, a dominant gene.**
Striatum（煞車）退化 → Huntington，顯性基因。

**Key:** Caudate + putamen = striatum = brakes.
判斷關鍵：caudate＋putamen＝striatum＝煞車。

**A man felt flat and joyless for two years before a resting tremor started. Why did mood change first?** [Motor disorders (Huntington and Parkinson)]
先兩年憂鬱，後來才手抖。為什麼情緒先出現？

→ **Professor's version (10/22): substantia nigra dopamine also goes to the pleasure center, the nucleus accumbens, so the pleasure gas runs out first. (Textbook anatomy: the main input to the nucleus accumbens is from the VTA.)**
教授版本（10/22）：黑質的多巴胺也送到快樂中樞 nucleus accumbens，快樂的油門先沒了。（課本解剖：送到依核的主要是 VTA）

**Key:** Mood before tremor = dopamine to the reward center drops first. It is not the striatum suppressing the prefrontal cortex.
判斷關鍵：不是紋狀體去壓前額葉。

**A student sits completely still. What are the basal ganglia doing?** [Motor disorders (Huntington and Parkinson)]
學生坐著不動，基底核在做什麼？

→ **The gas and brakes are both working at full force, equally balanced (equilibrium).**
油門和煞車都在全力運作，力量相等（equilibrium）。

**Key:** Sitting still is not rest.
判斷關鍵：坐著不動不是休息。

**Cortex and muscle are normal, but motor axon signals are slow and out of step. Problem?** [Motor disorders (Huntington and Parkinson)]
皮質和肌肉都正常，軸突訊號又慢又散。問題？

→ **Demyelination (for example, MS).**
Demyelination（例如 MS）。

**Key:** Slow, out-of-step signals = myelin. Cross out whatever the question calls normal first; hierarchy: muscle, myelin, spinal cord, basal ganglia, cortex.
判斷關鍵：題目說正常的先劃掉。階層：肌肉 → 髓鞘 → 脊髓 → 基底核 → 皮質。

**Why does Parkinson often start on one side of the body?** [Motor disorders (Huntington and Parkinson)]
帕金森為什麼常從一側開始？

→ **Each side of the brain has its own basal ganglia, and neurons on one side may die first.**
兩側各有基底核，一側的神經元可能先死。

**Key:** "The pathway crosses at the medulla" is the right fact used in the wrong place.
判斷關鍵：交叉點在延腦這個說法用錯地方。

## 17. Mirror neurons （p.266）

**A musician sits motionless watching a pianist; his motor neurons fire in finger patterns. Which neurons?** [Mirror neurons]
音樂家坐著不動看鋼琴家，運動區神經元照手指模式放電。是什麼？

→ **Mirror neurons: they fire both when you act and when you watch someone else act.**
Mirror neurons：自己做和看別人做都放電。

**Key:** Motionless = his hands do not move, so it is not alpha motor neurons. The link to autism is only speculation, not proven.
判斷關鍵：motionless＝手沒動，所以不是 alpha 運動神經元。和自閉症的關聯只是推測，不是證實。

# 第九章 恆定、動機、獎賞（p.282–311）

## 18. Drive reduction vs reward, homeostasis, set point （Ch9 p.282–311）

**Sam eats plain crackers until hunger goes away; later, full, he orders chocolate cake. Difference?** [Drive reduction vs reward, homeostasis, set point]
餓了吃餅乾到不餓，飽了還點蛋糕。差在哪？

→ **Crackers = drive reduction (push); cake = reward (pull).**
餅乾＝drive reduction（推）；蛋糕＝reward（拉）。

**Key:** Eating until hunger is gone = reducing a need (push); eating when full = an incentive (pull). Both are motivation.
判斷關鍵：兩者都是動機。

**Which is the setting, and which is the concept: 37°C or keeping temperature stable?** [Drive reduction vs reward, homeostasis, set point]
37 度和維持體溫穩定，哪個是設定值、哪個是概念？

→ **37°C = set point; keeping temperature stable = homeostasis.**
37°C＝set point；維持穩定＝homeostasis。

**Key:** Professor 11/3: homeostasis is the concept; the set point is the setting.
判斷關鍵：教授 11/3：homeostasis 是概念，set point 是設定。

## 19. Body temperature, endotherm vs ectotherm, hyper- vs hypothermia （Ch9 p.282–311）

**A lizard basks before moving fast; a squirrel holds its temperature; a runner's core temperature climbs. Labels?** [Body temperature, endotherm vs ectotherm, hyper- vs hypothermia]
蜥蜴曬太陽、松鼠體溫穩定、跑者體溫飆高。各是什麼？

→ **Lizard = ectotherm; squirrel = endotherm; runner = hyperthermia.**
蜥蜴 ectotherm、松鼠 endotherm、跑者 hyperthermia。

**Key:** Basks in the sun = depends on the environment; holds its own temperature = makes its own heat. Hyper = too much, hypo = too little.
判斷關鍵：hyper 太多、hypo 太少。

## 20. Temperature-sensitive neurons of the hypothalamus （Ch9 p.282–311）

**Early flu: normal temperature, yet she shivers under blankets feeling freezing. Why?** [Temperature-sensitive neurons of the hypothalamus]
感冒初期體溫正常，卻蓋三條被子發抖。為什麼？

→ **Pyrogens raise the hypothalamus's set point, so a normal temperature is read as too cold.**
Pyrogens 把下視丘設定點調高，正常體溫被當成太冷。

**Key:** Shivering is a defense against cold; warm-sensitive neurons handle cooling.
判斷關鍵：發抖是冷的防禦；warm-sensitive 神經元負責散熱。

## 21. Fluids, osmosis, isotonic, hypotonic, hypertonic （Ch9 p.282–311）

**A patient gets an IV of a very dilute solution. What happens to his cells?** [Fluids, osmosis, isotonic, hypotonic, hypertonic]
點滴是很稀的溶液，細胞怎樣？

→ **Hypotonic: water moves into the cells, and they swell.**
Hypotonic：水進細胞，細胞脹大。

**Key:** Dilute outside = hypotonic = water moves in. Only hypertonic makes cells shrink.
判斷關鍵：外面稀＝低張＝水往裡跑。高張才縮水。

**Cells are placed in 3% saline. What happens?** [Fluids, osmosis, isotonic, hypotonic, hypertonic]
細胞放進 3% 鹽水，會怎樣？

→ **Hypertonic: water moves out of the cells, and they shrink.**
Hypertonic：水跑出來，細胞縮水。

**Key:** Turn the scenario into the term: 3% saline → hypertonic → shrink.
判斷關鍵：情境轉術語：3% saline → hypertonic → shrink。

## 22. Hypovolemic vs osmotic thirst, osmoreceptors （Ch9 p.282–311）

**Minutes after salty pretzels, with no sweating or blood loss, you're very thirsty. Sequence?** [Hypovolemic vs osmotic thirst, osmoreceptors]
吃完鹹餅乾很渴，沒流汗沒失血。順序？

→ **Hypertonic extracellular fluid → cells shrink → OVLT osmoreceptors → thirst + ADH.**
細胞外高張 → 細胞縮水 → OVLT 滲透壓受器 → 口渴＋ADH。

**Key:** Salty food with no blood loss = osmotic thirst, not hypovolemic.
判斷關鍵：沒失血＝不是低血容量。

**A soldier loses a lot of blood and is thirsty; he gets only plain water. Which is correct?** [Hypovolemic vs osmotic thirst, osmoreceptors]
士兵失血很渴，只給白開水。哪個對？

→ **Hypovolemic thirst (baroreceptors, kidneys, angiotensin II → subfornical organ); he needs water plus salt.**
Hypovolemic thirst（baroreceptors、腎臟、angiotensin II → subfornical organ）；要補水＋鹽。

**Key:** Blood loss removes water and salt together, so plain water dilutes the blood. The two kinds of thirst feel the same.
判斷關鍵：失血是水和鹽一起少；只補水會把血稀釋。兩種口渴感覺一樣。

## 23. Glucose, glucoreceptors, insulin and glucagon （Ch9 p.282–311）

**Hours into a marathon without food, her blood glucose falls but her brain still gets glucose. Which response?** [Glucose, glucoreceptors, insulin and glucagon]
馬拉松沒吃東西，血糖下降，大腦還有葡萄糖。哪個反應？

→ **The pancreas releases glucagon, which turns glycogen back into glucose.**
Pancreas 放 glucagon，肝醣變回葡萄糖。

**Key:** Low blood sugar = glucagon; insulin stores glucose (swap trap).
判斷關鍵：血糖低＝升糖素；胰島素是存起來（對調陷阱）。

**Right after a big meal, which hormone moves glucose into cells?** [Glucose, glucoreceptors, insulin and glucagon]
大吃一頓後，哪個荷爾蒙把葡萄糖送進細胞？

→ **Insulin.**
Insulin。

**Key:** High blood sugar = insulin.
判斷關鍵：血糖高＝胰島素。

## 24. Fat, leptin, ghrelin, orexins, feeding （Ch9 p.282–311）

**Before lunch, her stomach is empty and a bakery smell makes her hungry. Mechanism?** [Fat, leptin, ghrelin, orexins, feeding]
午餐前空腹，聞到麵包香很餓。機制？

→ **Ghrelin (stomach) → arcuate nucleus → orexin neurons → hunger.**
Ghrelin（胃）→ 弓狀核 → orexin 神經元 → 想吃。

**Key:** Empty stomach = ghrelin; leptin is fat saying "enough" (swap trap).
判斷關鍵：leptin 是脂肪說夠了（對調陷阱）。

**Leptin-less mice overeat and leptin shots help them, yet many obese people have high leptin and stay hungry. Why?** [Fat, leptin, ghrelin, orexins, feeding]
缺 leptin 的老鼠打了會少吃，但胖的人 leptin 高還是餓。為什麼？

→ **Leptin from fat tells the hypothalamus that stores are full; many obese people develop leptin resistance.**
Leptin 從脂肪告訴下視丘存夠了；很多胖的人產生阻抗。

**Key:** High leptin but still hungry = resistance. The hunger signal from the stomach is ghrelin.
判斷關鍵：飢餓訊號從胃來的是 ghrelin。

## 25. Reward circuits, cocaine （Ch9 p.282–311）

**A drug blocks dopamine transporters in the nucleus accumbens. Effect?** [Reward circuits, cocaine]
藥物擋住依核的多巴胺轉運體，效果？

→ **Dopamine stays in the synapse longer, strengthening VTA → NAc; the drug is an agonist.**
多巴胺留在突觸更久，強化 VTA → NAc，是 agonist。

**Key:** It blocks reuptake, not the receptor, so the effect gets stronger.
判斷關鍵：擋的是回收，不是受體，所以作用變強。

**A rat always got a drug in a green box; the same dose in a red box killed it. Why?** [Reward circuits, cocaine]
老鼠一直在綠盒拿藥，紅盒同劑量死了。為什麼？

→ **The brain prepares in advance based on context; without the familiar cue there is no preparation, so the same dose becomes an overdose.**
大腦根據情境預先準備；沒有熟悉線索就沒準備，同劑量變過量。

**Key:** Same dose, new box = no learned preparation. Professor 10/27 class: addiction is a disease, not a choice.
判斷關鍵：教授 10/27 課堂。成癮是疾病，不是選擇。

## 第十章 性別與性分化（p.314 起）

### 26. Sex vs gender, 4 steps of sexual differentiation （p.314, 317–318, 320, 324）

**An XY person's SRY gene doesn't work. Ducts, genitals, brain?** [Sex vs gender, 4 steps of sexual differentiation]
XY 但 SRY 不作用。管道、外生殖器、大腦？

→ **Müllerian ducts, female genitals, female brain: fully female.**
Müllerian、女性、女性：完全女性。

**Key:** SRY is the first domino; the main job of the Y chromosome is SRY.
判斷關鍵：SRY 是第一張骨牌；Y 的主要工作就是 SRY。

**Why doesn't a pregnant mother's high estradiol masculinize her XX fetus's brain?** [Sex vs gender, 4 steps of sexual differentiation]
媽媽雌二醇很高，為什麼不會讓 XX 胎兒大腦男性化？

→ **Alpha-fetoprotein blocks estradiol in the blood; testosterone becomes estradiol through aromatase only after it enters neurons.**
Alpha-fetoprotein 擋住血中的雌二醇；睪固酮進神經元後才被 aromatase 轉成雌二醇。

**Key:** The brain is masculinized by estradiol (aromatase); the external genitals by DHT.
判斷關鍵：大腦靠 estradiol（aromatase）；外生殖器靠 DHT。

**An XY boy has plenty of testosterone but no aromatase in his neurons. Brain?** [Sex vs gender, 4 steps of sexual differentiation]
XY 男孩睪固酮很多，但神經元沒有 aromatase。大腦？

→ **A fully female brain.**
完全女性。

**Key:** Professor 11/3: testosterone is only a pass; it must become estrogen inside the cell to masculinize the brain.
判斷關鍵：教授 11/3：睪固酮只是通行證，要在細胞裡變成雌激素才男性化。

**A character has a male body, identifies as a woman, and is attracted to women. What point?** [Sex vs gender, 4 steps of sexual differentiation]
男性身體、認同女性、喜歡女性。說明什麼？

→ **Biological sex, gender identity, and sexual orientation are three separate things.**
生理性別、性別認同、性傾向是三件分開的事。

**Key:** "Always" and "determines" are absolute words, so those options are wrong.
判斷關鍵：always、determines 是絕對字。

**What triggers puberty in both sexes?** [Sex vs gender, 4 steps of sexual differentiation]
什麼啟動青春期（男女都有）？

→ **The ratio of body fat to height.**
脂肪和身高的比例。

**Key:** Professor 11/3 version. SRY, MIH, and TDF exist only in males, but females also go through puberty, so cross them out.
判斷關鍵：教授 11/3 版本。刪去法：SRY、MIH、TDF 只有男性有，女性也有青春期。所以初經越來越早、飲食失調可能不來。

## 27. Sexual dimorphism of the brain, sexual orientation （p.325–327, 334–336）

**INAH-3 is larger in heterosexual men than women and smaller in gay men. Best conclusion?** [Sexual dimorphism of the brain, sexual orientation]
INAH-3 異性戀男性比女性大，同性戀男性比較小。最好的結論？

→ **Some hypothalamic nuclei are sexually dimorphic, and their size is associated with sexual orientation.**
有些下視丘核是性別雙型的，大小和性傾向有關聯。

**Key:** "Proves" and "solely" are too strong; a correlation does not mean it is the only cause.
判斷關鍵：proves、solely 太強；相關不等於唯一原因。

## 28. Sexual differentiation irregularities （p.317–323）

**XY, testes make testosterone, but androgen receptors don't respond (AIS). Development?** [Sexual differentiation irregularities]
XY 有睪丸會分泌睪固酮，但受體沒反應。發育？

→ **No ducts of either type, a female appearance, and normal female puberty (breasts develop, infertile).**
兩套管道都沒有、外觀女性、青春期正常女性發育（有胸部、不能生育）。

**Key:** Testes = MIH removes the Müllerian ducts; testosterone has no effect = Wolffian ducts do not develop. Confirmed by the professor 11/3.
判斷關鍵：有睪丸＝有 MIH 殺掉 Müllerian；睪固酮沒作用＝Wolffian 長不出來。教授 11/3 確認。

**XY, normal testosterone and MIH, but no 5-alpha reductase. At birth?** [Sexual differentiation irregularities]
XY 睪固酮和 MIH 正常，缺 5-alpha 還原酶。出生時？

→ **Wolffian ducts, female-looking genitals, masculinized brain; masculinization begins at puberty.**
Wolffian、外觀女性、大腦男性化；青春期開始男性化。

**Key:** 5-alpha reductase only controls DHT for the penis: a perfect male except for the penis.
判斷關鍵：5-alpha 只管 DHT 長陰莖：perfect male except no penis。

**XX fetus exposed to large amounts of testosterone. Ducts, genitals, brain?** [Sexual differentiation irregularities]
XX 胎兒受到大量睪固酮。管道、外生殖器、大腦？

→ **Both duct systems, masculinized genitals, masculinized brain.**
兩套都有、男性化、男性化。

**Key:** Class activity version (11/3). No testes = no MIH, so Müllerian ducts still grow; testosterone works the same wherever it comes from, and genitals depend on the timing window.
判斷關鍵：課堂活動版本（11/3）。沒睪丸＝沒 MIH，Müllerian 照長；睪固酮從哪來都一樣。外生殖器要對到時間窗口。

**Testosterone present but no working MIH (PMDS). Ducts?** [Sexual differentiation irregularities]
有睪固酮但 MIH 沒作用。管道？

→ **Both duct systems; male genitals; masculinized brain.**
兩套都有；外生殖器男性、大腦男性化。

**Key:** First ask whether there are testes (MIH); then ask whether testosterone works (Wolffian).
判斷關鍵：先問有沒有睪丸（MIH），再問睪固酮有沒有作用（Wolffian）。

## 29. Genetic abnormalities with X and Y （p.315–316）

**A tall man with small testes and low testosterone has XXY. Which is correct?** [Genetic abnormalities with X and Y]
高個子、睪丸小、睪固酮低，XXY。哪個對？

→ **He is male because he has SRY; the extra X lowers his testosterone (Klinefelter syndrome).**
有 SRY 所以是男性；多一條 X 讓睪固酮偏低（Klinefelter）。

**Key:** The X does not outvote the Y; SRY present = testes develop.
判斷關鍵：X 不會投票贏過 Y；有 SRY 就長睪丸。

## 30. Hormonal and genetic effects on development, gender differences （p.320–328）

**An XX girl exposed to high adrenal androgens before birth prefers rough play and scores high on mental rotation. Why?** [Hormonal and genetic effects on development, gender differences]
XX 女孩出生前雄性素很高，喜歡粗野遊戲、空間能力好。為什麼？

→ **Prenatal androgens had an organizational effect that shaped her brain (CAH).**
出生前雄性素的 organizing 作用塑造了大腦（CAH）。

**Key:** Appears in childhood = before puberty, so it is not the activational effect.
判斷關鍵：童年就出現＝不是青春期的 activational。

## 31. Regulation of sex hormones in adults, parental behavior （p.337, 340, 342–343）

**A bodybuilder takes high doses of testosterone; his testes shrink. Why?** [Regulation of sex hormones in adults, parental behavior]
健美選手打大量睪固酮，睪丸萎縮。為什麼？

→ **Negative feedback lowers GnRH, then LH and FSH, so the testes are not stimulated.**
負回饋壓低 GnRH → LH、FSH，睪丸沒被刺激。

**Key:** Direction: lowered, not raised.
判斷關鍵：方向：壓低，不是調高。

**Blocking vasopressin in voles' reward areas stops pair bonding; men's testosterone drops when they become fathers. Conclusion?** [Regulation of sex hormones in adults, parental behavior]
擋住田鼠獎賞區的 vasopressin 就不配對；男人當爸爸睪固酮下降。結論？

→ **Vasopressin and oxytocin support pair bonding and parenting; testosterone drops when men become fathers.**
Vasopressin、oxytocin 支持配對和育兒；當爸爸睪固酮下降。

**Key:** Direction: drops, not rises.
判斷關鍵：方向：下降，不是上升。

# 補充頁：外部題組比對後新增（11/4 晚）

四家 AI 比對 Brainscape、Knowt 題組後，只收「對得到 Study Guide、小抄沒有、跟教授版本不衝突、有原文證據」的內容。跟教授版本衝突時，一律以前面 38 頁為準。

| | |
|---|---|
| **Ch7**<br>SG5 | **Auditory nerve = axons of spiral ganglion neurons (p.215 right col., last paragraph).**<br>聽神經＝螺旋神經節神經元的軸突（p.215 右欄最後一段）。 |
| **Ch7**<br>SG5 | **Full hearing path: cochlea → auditory nerve → cochlear nuclei (medulla) → superior olive → inferior colliculus → MGN (thalamus) → A1 (temporal lobe). Fig. 7.7.**<br>完整聽覺路徑：耳蝸 → 聽神經 → 延腦耳蝸核 → 上橄欖核 → 下丘 → MGN（視丘）→ 顳葉 A1。圖 7.7。 |
| **Ch7**<br>SG8 | **Vestibular system has two parts: otolith organs (utricle, saccule) sense head tilt and straight-line movement; semicircular canals sense rotation.**<br>前庭有兩部分：耳石器（橢圓囊、球囊）感覺頭的傾斜和直線移動；半規管感覺旋轉（轉圈頭暈）。 |
| **Ch7**<br>SG7 | **Nociceptors are free nerve endings. A-delta fibers = fast, sharp pain; C fibers = slow, dull, aching pain.**<br>痛覺受器是游離神經末梢。A-delta 纖維＝快、尖銳的痛；C 纖維＝慢、悶痛。 |
| **Ch7**<br>SG10 | **Gate theory was proposed by Melzack and Wall (1965): a gate in the dorsal horn decides which signals reach the brain.**<br>閘門理論是 Melzack 和 Wall（1965）提出的：脊髓背角的門決定哪些訊號上到大腦。 |
| **Ch8**<br>SG13 | **Stretch reflex = the knee jerk (muscle spindle, monosynaptic). Same thing, two names.**<br>Stretch reflex（牽張反射）就是膝跳反射，同一件事兩個名字。 |
| **Ch8**<br>SG13 | **Lateral corticospinal tract: crosses at the medulla (pyramids), controls fine limb and finger movement. Ventral (medial) tract: trunk muscles.**<br>外側皮質脊髓徑：在延腦錐體交叉，管四肢、手指的精細動作。腹側（內側）徑：管軀幹。 |
| **Ch8**<br>SG16 | **If an option says the globus pallidus is the brake, answer with Dr. Keller's version: substantia nigra = gas, striatum = brakes.**<br>選項如果說 globus pallidus 是煞車，照教授版本作答：黑質＝油門、紋狀體＝煞車。 |
| **Ch6**<br>SG4 | **V1 = primary visual cortex = striate cortex (occipital). Three names, one place.**<br>V1＝初級視覺皮質＝紋狀皮質（枕葉），三個名字同一個地方。 |
| **Ch9**<br>SG21 | **Extracellular fluid is high in Na+ and Cl-; intracellular fluid is high in K+.**<br>細胞外液鈉、氯多；細胞內液鉀多。 |
| **Ch9**<br>SG22 | **ADH (vasopressin) is released from the posterior pituitary; kidneys make less urine.**<br>ADH（vasopressin）由腦下垂體後葉釋放，腎臟減少尿量。 |
| **Ch9**<br>SG25 | **VTA → nucleus accumbens = mesolimbic pathway (runs in the medial forebrain bundle). Most addictive drugs act as dopamine agonists there.**<br>VTA → 依核＝中腦邊緣路徑（走在內側前腦束裡）。多數成癮藥物在這裡是多巴胺 agonist。 |
| **Ch10**<br>SG26 | **SRY switches on around week 6 of gestation; its protein (TDF) turns the gonads into testes.**<br>SRY 大約在懷孕第 6 週啟動，它的蛋白（TDF）把性腺變成睪丸。 |
| **Ch10**<br>SG29 | **XYY: male (has SRY). Same logic as XXY: a Y with SRY = testes = male.**<br>XYY：男性（有 SRY）。跟 XXY 同一個邏輯：有帶 SRY 的 Y＝睪丸＝男性。 |
| **Ch10**<br>SG30 | **Males get a second, short androgen surge at about 1 to 3 months after birth (mini-puberty).**<br>男嬰出生後大約 1 到 3 個月有第二次短暫的雄性素高峰（mini-puberty，小青春期）。 |